Multigrid for the even-odd preconditioned Möbius domain-wall operator with a factorized Pauli–Villars coarse correction

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Domain-wall fermions keep chiral symmetry up to effects that vanish as L_s grows, but inverting the Möbius domain-wall operator suffers from critical slowing down toward the physical light-quark mass. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator and dividing out the site-diagonal part, $(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, moves all eigenvalues into the right half-plane and compresses the high end of the spectrum, so that Krylov methods apply; earlier Pauli–Villars multigrid methods did not outperform red-black CG in four dimensions, as Boyle and Yamaguchi found.

In this work, we construct a two-grid method for its even-odd form in which the quark and Pauli–Villars factors are coarsened separately. Every aggregate contains the whole fifth dimension so that the coarse lattice is four-dimensional with nearest-neighbor couplings, and levels below the coarse lattice replace deflation. These choices address the three reasons Boyle and Yamaguchi gave for why this approach did not beat red-black CG. A single setup serves every value of Neff's α. The field of values of the preconditioned operator reaches into the left half-plane, yet at $\alpha = 1$ its spectrum has $\mathrm{Re}\,\lambda \geq 0.70$, and on 12^4 (one configuration) its pseudospectra bound the convergence of full GMRES. On four quenched $24^3 \times 48$ configurations the iteration count grows by a factor of 2.8–5.4 from $m = 0.1$ to $m = 0.001$, against 20–32 for mixed-precision even-odd CGNE. At $m = 0.001$ a solve is 4.1–7.3 times faster than mixed-precision CGNE, and a point-source propagator takes 0.40–0.56 of its time, setup included. For future HMC work, which needs solves with $D^{\dagger}$, we derive an exact relation, $D^{\dagger}\Gamma_5 D^{-1}\Gamma_5 = \Gamma_5(1-N)\Gamma_5$ (with $\Gamma_5 = \gamma_5 R$, R the reflection of the fifth dimension, and N proportional to the Möbius coefficient c). It makes $\Gamma_5 D^{-1}\Gamma_5$ a preconditioner for these solves without a second multigrid setup.

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I. Introduction

Many lattice computations need fermions whose chiral symmetry is broken only slightly at finite lattice spacing. Operators of different chirality then do not mix, which weak matrix elements require, as in the Standard Model calculation of $K \to \pi\pi$ decays and ε'/ε [1]. Precision calculations at the physical quark masses use such fermions as well: the hadronic vacuum polarization and light-by-light contributions to the muon $g-2$ [2, 3, 4], the $B \to \pi\ell\nu$ form factors on fine lattices with the heavy quark in the same Möbius domain-wall action [5], inclusive semileptonic B decays [6, 7, 8], and parton distribution amplitudes [9].

The overlap operator [10, 11] is exactly chiral on the lattice. Domain-wall fermions [12, 13, 14] cost less: they add a fifth dimension of extent L_s, in which a large negative Wilson mass $-M_5$ binds modes of opposite chirality to its two boundaries, and the residual breaking falls as L_s grows. The Möbius generalization [15, 16, 17] adds two coefficients (b, c) that reduce the residual breaking at fixed L_s; $c = 0$ is Shamir's operator.

Solving with the domain-wall operator is expensive, because it is neither Hermitian nor positive and one application costs L_s four-dimensional Wilson hops. The standard solver is still the conjugate gradient method [21] on the normal equation of the even-odd operator (CGNE), whose condition number is the square of that of the operator. As the quark mass m decreases, the smallest eigenvalues of D approach m, the condition number of the normal equation grows like $1/m^2$ and the CG iterations like $1/m$ until the finite volume limits them: critical slowing down. For Wilson fermions, multigrid removed it. The low modes that slow down Krylov solvers are locally coherent [20]: a few of them, cut into blocks, span the low-mode space. Adaptive aggregation multigrid uses this, and its cost hardly depends on the quark mass [27, 28, 29, 30, 31, 32].

Multigrid for domain-wall fermions. For domain-wall fermions multigrid is harder. Because of the large negative Wilson mass, the spectrum encloses the origin. Non-Hermitian Krylov methods then converge only after a number of steps of the order of the system size [35, 36]. Boyle [36] lists four approaches that work to some extent. They differ in the operator that is coarsened.

- Cohen *et al.* [37] coarsened the squared operator, with blocks that cover the whole fifth dimension. Its coarse operator reaches two hops.
- HDCG [38] accelerates CGNE and coarsens the squared even-odd Schur complement. Its coarse stencil has 81 points for blocks of at least 4^4.
- HDCR [39] coarsens the Hermitian indefinite operator $\gamma_5 R D$ at $c = 0$, where R reflects the fifth dimension. Its coarse operator stays nearest-neighbor, which keeps the setup affordable in HMC.
- Brower, Clark, Howarth and Weinberg [41] noted that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$ carries the physical four-dimensional degrees of freedom. They solved $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ on all lattice sites. Projecting the Wilson kernel slice by slice gives coarse operators $\hat{D}$ and $\hat{D}_{\mathrm{PV}}$, both nearest-neighbor, and the coarse correction solves $\hat{D}_{\mathrm{PV}}^{\dagger} \hat{D}\, \hat{e} = \hat{r}$. They tested Shamir fermions in the two-dimensional Schwinger model.

Boyle and Yamaguchi [40] compared these approaches for Shamir fermions; on large volumes HDCR was set up and solved faster than a single even-odd CG solve. They also carried the Pauli–Villars method of Ref. [41] to four-dimensional QCD, where it needed about half the fine-matrix applications of red-black CG but 18–31 times its solve time. They gave three reasons. The relevant baseline is red-black CG, against which the full-lattice solve is slow. The coarse space keeps a fifth dimension of extent L_s, so its cost removes the gain. And reducing that cost would need further levels or coarse deflation, which the PV method did not have. They also found that CG converges faster on the Schur complement than on its left-preconditioned form (for the Möbius setups of this work at $\alpha = 0.6$ the order is reversed, Appendix D), and that an even-odd smoother did not work with the HDCR correction.

Recent work by Boyle and collaborators. The multiple right-hand-side HDCG of Ref. [36] solves the normal equation of the even-odd operator with a two-level CG whose coarse space is spanned by blocked low modes. Treating many right-hand sides at once makes the coarse space cheap, with gains of up to twenty over even-odd CG at physical quark mass. Boyle and Oare [43] coarsen the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$, a 33-point stencil, aiming at solves on a new gauge field as in HMC. Its field of values can reach the left half-plane although its spectrum does not, so the GCR bound of Eisenstat, Elman and Schultz [44] is lost; they add a shift to the smoothers. On $48^3 \times 96$ at the physical quark mass they report 578 fine-matrix applications, against 26500 for red-black CG.

Setup cost in ensemble generation. A multigrid setup is paid once per gauge field, which is affordable for propagators but not for HMC. Eick and Mawhinney [52] avoid the setup by blocking a fine Möbius ensemble to a coarse lattice by renormalization-group methods. The interpolated correction then carries high modes that the fine operator amplifies by four orders of magnitude, and they remove them with a Chebyshev filter of degree 150–300.

The preconditioned operator. Kanamori, Chen and Matsufuru [45] looked for a preconditioned Möbius operator local enough for a multigrid. Approximating D_{PV} by its block-diagonal part B_{PV} brings back eigenvalues with negative real part, so the Pauli–Villars factor has to be kept whole. Dividing out the site-diagonal part C as well gives $A' = (D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, whose eigenvalues all lie in the right half-plane, with the smallest condition number among the operators they studied. Neff's form of the domain-wall matrix [46, 47] rescales the fifth dimension by a site-local factor Q and changes the conditioning of the even-odd system.

This work. We construct a two-grid method for A' that takes up the Pauli–Villars route of Refs. [41, 40] and addresses the three reasons of Ref. [40] in turn. It is an adaptive aggregation multigrid [27, 30, 32]: the coarse space comes from computed near-null vectors cut into aggregates, and the coarse operators are Galerkin projections. First, the outer system is the even-odd one, as for red-black CG; its inverse contains the inverses of the quark operator and of the adjoint Pauli–Villars operator, and we replace each by a Galerkin inverse. Second, every aggregate contains the whole fifth dimension, so the coarse lattice is four-dimensional and both coarse operators are nearest-neighbor. Third, levels below the coarse lattice with a dense LU at the bottom take the place of deflation for the quark factor (Sec. III.B). The Pauli–Villars coarse system is solved by CG on a Tikhonov-regularized normal equation, which removes small singular values that the Galerkin projection creates. The prolongation lifts each test vector to all lattice sites in two ways, one for each coarse solve, which makes the correction exact on them for the right form (Eq. (3.7)). The smoother is a fixed Chebyshev polynomial, and the setup combines Chebyshev filtering with a bootstrap.

The second result concerns HMC, which solves with $D^{\dagger}$ as often as with D. We derive an exact relation, Eq. (4.2), that makes the multigrid of D, placed between two $\Gamma_5 = \gamma_5 R$, a preconditioner for $D^{\dagger}$, with no second setup. We test the method on quenched configurations from 12^4 to $24^3 \times 48$ on a single GPU against even-odd CGNE.

II. The operator and its preconditioning

A. Spectra of preconditioned Möbius operators

Where the spectrum lies decides which iterative methods can work. A Krylov method builds a residual polynomial p with $p(0) = 1$ that should be small on the spectrum. Such a polynomial of low degree exists only if a region that excludes the origin, such as an ellipse, encloses the spectrum. The Chebyshev smoother of Sec. III.D, defined by an ellipse in the right half-plane, needs the same. The eigenvalues of the domain-wall operator lie on both sides of the origin, and the preconditioners below move them to one side. Figures 1 and 2, from Ref. [45], show the spectrum without and with both preconditioners; Fig. 9 in Appendix B shows the other preconditioners discussed below.

The unpreconditioned operator. The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$, and so does the spectrum on a $2+1$-flavor configuration at the physical point (Appendix B, Fig. 9). So CG and BiCGStab [22] do not work on D itself, and the practical choice has been CG on a positive definite squared operator.

Site-diagonal preconditioning. The site-diagonal part C has an inverse that is the same matrix on every site (Sec. II.B.3). With it the spectrum of DC^{-1} shrinks, but it still lies on both sides of the origin.

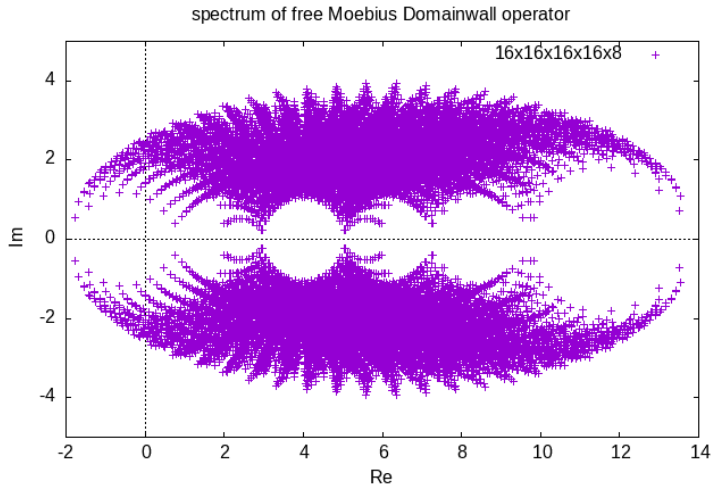


FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass 0.01, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. From Ref. [45], Fig. 1.

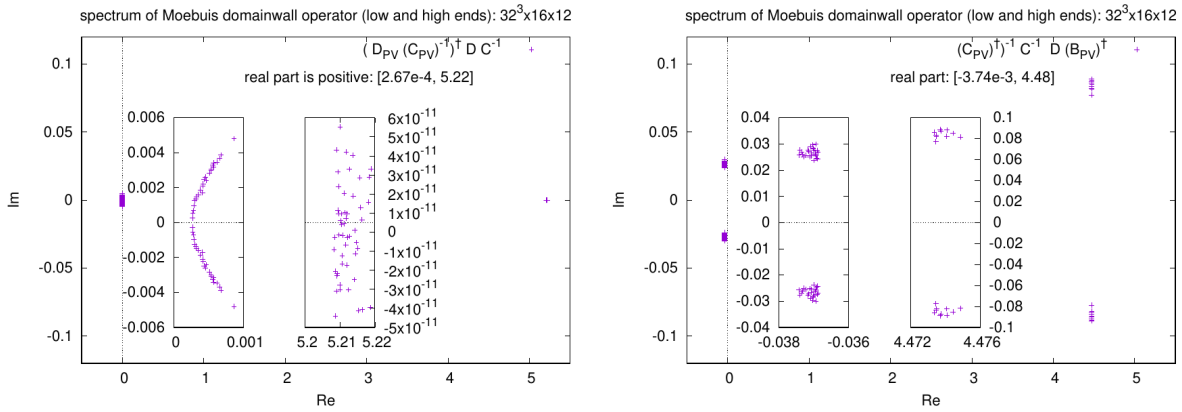


FIG. 2. Left: spectrum of $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$, Eq. (2.19), on a JLQCD $2+1$-flavor configuration at the physical point; all real parts are positive. Right: the same with D_{PV} replaced by its block-diagonal approximation B_{PV}; negative real parts reappear. The text of Ref. [45] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. From Ref. [45], Fig. 4.

Pauli–Villars preconditioning. The operator $D_{\mathrm{PV}}^{-1} D$, with $D_{\mathrm{PV}} = D(m = 1)$, approximates the overlap operator [41]. Since D_{PV}^{-1} costs as much as a solve with D, Ref. [41] used $D_{\mathrm{PV}}^\dagger$ instead. $D_{\mathrm{PV}}^\dagger D$ has positive real parts. Its low modes lie close to the imaginary axis, shifted by about the quark mass, as for an overlap operator. Its high end, however, reaches far beyond that of an overlap operator.

Both together. The operator $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}$ has all its eigenvalues in the right half-plane (Fig. 2, left), with the smallest condition number in Ref. [45]. With D_{PV} replaced by its block-diagonal approximation B_{PV}, which drops the hops between 4^4 blocks, negative real parts come back (right). Our method therefore keeps the factor exact on the fine lattice and approximates it only through its Galerkin coarse operator (Sec. III.A).

Two caveats. First, the figures show the full-lattice operator A'; the even-odd operators of Sec. II.C are not shown. Second, eigenvalues in the right half-plane do not give a field of values there. For a non-normal operator the GCR bound [44] needs the Hermitian part $(A + A^\dagger)/2$ to be positive definite, which can fail while every eigenvalue has positive real part, as Ref. [43] discusses for $D_{\mathrm{PV}}^\dagger D$. It fails for the even-odd form and for the preconditioned operator (Sec. V.D.1).

B. The operator and Pauli–Villars preconditioning

The notation follows Ref. [17] except where stated; Appendix A relates it to Ref. [45] and to Bridge++ symbol by symbol.

1. The Möbius operator

Ingredients. ψ is a field on the four-dimensional lattice with an extra index $s = 0, \dots, L_s - 1$, the fifth dimension; ψ_s is an ordinary Dirac field for each s. $P_\pm = \frac{1}{2}(1 \pm \gamma_5)$ are the chiral projectors. The Wilson hopping term is

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu) U_\mu(x) \psi(x+\hat{\mu}) + (1+\gamma_\mu) U_\mu^\dagger(x-\hat{\mu}) \psi(x-\hat{\mu}) \Big], \tag{2.1}$$

and the Wilson operator with mass $-M_5$ (the domain-wall height $M_5 > 0$) is

$$W = (4 - M_5) - \tfrac{1}{2} K. \tag{2.2}$$

This is the usual $D_W(-M_5)$ with hopping coefficient $\frac{1}{2}$; in the κ normalization $W = (4 - M_5)(1 - \kappa K)$ with $\kappa = 1/(8 - 2M_5)$. W acts on each slice s in the same way and does not mix slices. Its free spectrum, needed in Sec. IV, is $\{-M_5 + \sum_\mu (1 - \cos p_\mu) \pm i(\sum_\mu \sin^2 p_\mu)^{1/2}\}$.

The operator. With

$$D_+ = 1 + bW, \qquad D_- = cW - 1 \tag{2.3}$$

(Ref. [17], Eq. (2.18)), the Möbius operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+ \psi_s + D_- \big(P_- \psi_{s+1} + P_+ \psi_{s-1} \big), \qquad \psi_{L_s} \equiv -m\, \psi_0, \quad \psi_{-1} \equiv -m\, \psi_{L_s - 1} \tag{2.4}$$

(Ref. [17], Eq. (2.20), with s counted from 0). For $L_s = 4$, as a matrix in s,

$$D = \begin{pmatrix} D_+ & D_- P_- & 0 & -m D_- P_+ \\ D_- P_+ & D_+ & D_- P_- & 0 \\ 0 & D_- P_+ & D_+ & D_- P_- \\ -m D_- P_- & 0 & D_- P_+ & D_+ \end{pmatrix}. \tag{2.5}$$

Shamir's operator is $b = 1$, $c = 0$; the results of this work use $b = 1.5$, $c = 0.5$ ($b - c = 1$, $b + c = 2$). The quark mass enters only through the two corners. The physical four-dimensional quark field is $q = P_- \psi_0 + P_+ \psi_{L_s - 1}$.

The hop in the fifth dimension. Eq. (2.4) separates into a part that hops in the fifth dimension and one that does not. Define

$$(Y\psi)_s = -P_- \psi_{s+1} - P_+ \psi_{s-1} \qquad \text{(with the same wrap-around)}, \tag{2.6}$$

For $L_s = 4$, as a matrix in s,

$$Y = \begin{pmatrix} 0 & -P_- & 0 & m P_+ \\ -P_+ & 0 & -P_- & 0 \\ 0 & -P_+ & 0 & -P_- \\ m P_- & 0 & -P_+ & 0 \end{pmatrix}. \tag{2.7}$$

This is the off-diagonal part of Eq. (2.5) with the factor D_- removed and the sign reversed. Then $D = D_+ - D_- Y$. Here D_- stands to the *left* of Y, since in Eq. (2.4) it acts on the shifted and projected field, and this order matters in Sec. II.B.2. Inserting Eqs. (2.2) and (2.3),

$$D = 1 + bW - (cW - 1)Y = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2} K\,(b - cY), \qquad C = d_+ + d_- Y, \tag{2.8}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. In the last step $W = (4 - M_5) - \frac{1}{2}K$ was split into its site-local constant and its hop; the constant parts form C. Note $d_- = -(c(4 - M_5) - 1)$ is *minus* the site-local part of D_-, because Y carries the sign. The Pauli–Villars operator is the same operator at unit mass, $D_{\mathrm{PV}} = D|_{m=1}$, $C_{\mathrm{PV}} = C|_{m=1}$; $D_\pm$ and K do not depend on m.

2. Γ_5-transformation and the exact adjoint

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$, so a solve with $W^\dagger$ is a solve with W between two multiplications by γ_5. The analogue for domain-wall fermions uses

$$\Gamma_5 = \gamma_5 R, \qquad (R\psi)_s = \psi_{L_s-1-s}. \tag{2.9}$$

Each building block transforms as follows.

1. W, D_+, D_-: they act on every slice alike, so they commute with R, and $\gamma_5 W \gamma_5 = W^\dagger$. Hence $\Gamma_5 W \Gamma_5 = W^\dagger$, $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

2. Y: it has no hop in the four dimensions, so $\Gamma_5 Y \Gamma_5$ can be read off Eq. (2.6). γ_5 commutes with $P_\pm$; R turns the hop $s \to s+1$ into $s \to s-1$. So $\Gamma_5 Y \Gamma_5$ has $-P_-$ on the hop down and $-P_+$ on the hop up, and the corner $-m$ moves to the other corner. This is the transpose of Eq. (2.6) in s with $P_\pm$ in place, which is $Y^\dagger$ since $P_\pm^\dagger = P_\pm$ and m is real: $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

3. W and Y do not commute: γ_μ in K anticommutes with γ_5, so $\gamma_\mu P_\pm = P_\mp \gamma_\mu$, and the hop of K exchanges the two projectors of Y.

From $D = D_+ - D_- Y$ it follows that

$$\Gamma_5 D \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \qquad \text{whereas} \qquad D^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger, \tag{2.10}$$

the adjoint reversing the order of the product. The difference is a commutator,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger = c\,[Y^\dagger, W^\dagger] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big], \tag{2.11}$$

where $D_-^\dagger = cW^\dagger - 1$ and $W^\dagger = (4 - M_5) - \frac{1}{2}K^\dagger$ were used. It vanishes for Shamir's operator ($c = 0$) and is of order c otherwise; multigrid on the Γ_5-Hermitian operator has been used in the $c = 0$ subspace of Möbius fermions (as summarized in Ref. [36]). Thus $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for Möbius fermions: a multigrid built for D cannot be used for $D^\dagger$ by conjugating with Γ_5. On configuration 1 of 12^4 with stout links ($M_5 = 1$, $m = 0.001$), $\|\Gamma_5 D \Gamma_5 b - D^\dagger b\| / \|D^\dagger b\| = 0.15$ for $(b, c) = (1.5, 0.5)$, 0.22 for $(2, 1)$ and 0 for $(1, 0)$, for a random b.

The exact relation. It is the generalized Γ_5-Hermiticity of Ref. [17], Eqs. (2.22)–(2.23), whose Γ_5 contains D_-^{-1}; here $\Gamma_5 = \gamma_5 R$ and D_- appears explicitly. D_+ and D_- are functions of the same W and commute. Where D_- is invertible, factor it out to the left:

$$D = D_-\, K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{2.12}$$

K_D is Γ_5-Hermitian: by item 1, $\Gamma_5 D_-^{-1} D_+ \Gamma_5 = (D_-^\dagger)^{-1} D_+^\dagger$, and this equals $(D_-^{-1} D_+)^\dagger = D_+^\dagger (D_-^\dagger)^{-1}$ because $D_+^\dagger$ and $D_-^\dagger$ commute; with item 2, $\Gamma_5 K_D \Gamma_5 = K_D^\dagger$. Then

$$D^\dagger = K_D^\dagger D_-^\dagger = \Gamma_5 K_D \Gamma_5\, D_-^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger, \tag{2.13}$$

and multiplying by $D_- \Gamma_5$ from the left,

$$\boxed{D_-\, \Gamma_5\, D^\dagger = D\, \Gamma_5\, D_-^\dagger} \tag{2.14}$$

This form has no inverse, holds for any m (so also for $D_{\rm PV}$, with the same D_-), and can be checked by applying both sides to a vector. On the configuration above both sides agree to 3×10^{-16} for $(b, c) = (1.5, 0.5)$ and $(2, 1)$. The Hermitian operator of the Möbius case is $\Gamma_5 K_D = \Gamma_5 D_-^{-1} D$; Ref. [17], Eq. (2.22), defines its Γ_5 as $\gamma_5 R D_-^{-1}$ for that reason. It contains D_-^{-1}, whose cost is discussed in Sec. IV.

3. The site-diagonal part and its inverse

$C = d_+ + d_- Y$ of Eq. (2.8) is what remains of D when the four-dimensional hops are dropped. It acts within one four-dimensional site, as an $L_s \times L_s$ matrix in s times the identity in color, and it commutes with $P_\pm$ (only Y is involved, and Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$). So it splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. From Eq. (2.6), Y moves ψ^+ from $s-1$ to s and ψ^- from $s+1$ to s, with the factor -1, and the wrap-around gives $+m$:

$$(C\psi)^+_s = \begin{cases} d_+\psi^+_0 + m\, d_-\psi^+_{L_s-1}, & s = 0, \\ d_+\psi^+_s - d_-\psi^+_{s-1}, & s \geq 1, \end{cases} \tag{2.15}$$

a lower bidiagonal matrix with one corner element; for ψ^- the same with $s \to L_s - 1 - s$ (upper bidiagonal). For $L_s = 4$ the block on ψ^+ is

$$C^+ = \begin{pmatrix} d_+ & 0 & 0 & m\, d_- \\ -d_- & d_+ & 0 & 0 \\ 0 & -d_- & d_+ & 0 \\ 0 & 0 & -d_- & d_+ \end{pmatrix}. \tag{2.16}$$

The inverse. To solve $C^+x = r$, write the rows $s \geq 1$ as $x_s = (r_s + d_- x_{s-1})/d_+$. Starting from an unknown x_0, each x_s is affine in x_0: with $q = d_-/d_+$,

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \quad p_s = (r_s + d_- p_{s-1})/d_+. \tag{2.17}$$

The row $s = 0$, $d_+x_0 + m\, d_- x_{L_s-1} = r_0$, with $x_{L_s-1} = p_{L_s-1} + q^{L_s-1}x_0$, fixes x_0:

$$x_0 = \frac{r_0 - m\, d_-\, p_{L_s-1}}{d_+ + m\, d_-\, q^{L_s-1}}, \qquad x_s = p_s + q^s x_0. \tag{2.18}$$

For C^- the substitution runs from $s = L_s - 1$ downwards. $C^\dagger$ has the transposed blocks: on ψ^+ an upper bidiagonal matrix, solved like C^-, and on ψ^- a lower one. C^{-1} is thus exact, costs $O(L_s)$ per site, and does not depend on the gauge field [45]. It exists unless $d_+ + m\, d_- q^{L_s-1} = 0$.

4. The preconditioned operator

Ref. [45] multiplies $D\psi = \eta$ from the left by $(D_{\rm PV}C_{\rm PV}^{-1})^\dagger$ and substitutes $\psi = C^{-1}y$:

$$A'y = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger\, \eta, \qquad A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger\, D\, C^{-1}, \qquad \psi = C^{-1}y. \tag{2.19}$$

The two factors are chosen for these reasons:

- For $m = 1$, $DC^{-1} = D_{\rm PV}C_{\rm PV}^{-1}$ and $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger(D_{\rm PV}C_{\rm PV}^{-1})$, which is Hermitian and positive definite. For $m < 1$ the two factors differ only in the corners of Eq. (2.5), and A' is a perturbation of a positive operator.
- $D_{\rm PV}^{-1}D$ approximates the overlap operator [41]; $D_{\rm PV}^\dagger$ in place of $D_{\rm PV}^{-1}$ keeps the cost of one application of A' at two applications of D plus the site-local C^{-1} and $C_{\rm PV}^{-1}$.

5. Link smearing

The fermion operator may be built from smeared links in place of $U_\mu(x)$; smearing integrates the surrounding gauge field into each link and suppresses the short-distance fluctuations that produce near-zero modes of W [18]. We use the stout projection combined with the APE staples [19]:

$$C_\mu(x) = \sum_{\nu\neq\mu} \rho\Big[U_\nu(x)U_\mu(x+\hat\nu)U_\nu^\dagger(x+\hat\mu) + U_\nu^\dagger(x-\hat\nu)U_\mu(x-\hat\nu)U_\nu(x-\hat\nu+\hat\mu)\Big], \tag{2.20}$$

$$U'_\mu(x) = \exp\Big(\big[C_\mu(x)U_\mu^\dagger(x)\big]_{\rm AT}\Big)\, U_\mu(x), \tag{2.21}$$

where $[\cdots]_{\rm AT}$ is the anti-Hermitian traceless part and ρ a tunable parameter. The step (2.21) is applied repeatedly; "stout $\rho \times n$" means n steps with the same ρ. Only W (through K) sees the smeared links; Y, C and Q do not depend on the gauge field.

6. *Neff's* α

Neff's form of the domain-wall matrix [46, 47] is $D^{(\alpha)} = DQ$ with a site-local Q, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \;\; (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{2.22}$$

The following properties, each read off Eq. (2.22), are used below:

- Q is Hermitian, and Q^{-1} is Eq. (2.22) with $\alpha \to 1/\alpha$.
- Q leaves the components $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field q; so $D^{(\alpha)}$ and D have the same four-dimensional propagator.
- The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$, since $D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} = D\psi$.
- Q is site-local, so the site-diagonal part of DQ is CQ, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$.

The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D_{\rm PV}^{(\alpha)} = D_{\rm PV}Q$ in the latter case. For $\alpha < 1$ it improves the conditioning of the even-odd operator [46, 47].

C. Even-odd forms

In the even-odd block form the diagonal blocks of D are C. Order the sites so that the even ones come first. We write X_{pq} for the block of an operator X that maps the sites of parity q to those of parity p, and ψ_e, ψ_o for the even and odd parts of a field. The hopping term K connects nearest neighbors, which always have opposite parity, so it has no even–even or odd–odd block. In $D = C - \frac{1}{2}K(b - cY)$ (Eq. (2.8)) the diagonal blocks therefore come from C alone and the off-diagonal ones from the hop alone:

$$D = \begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b - cY) \;\; (p \neq q), \qquad C = \begin{pmatrix} C_e & 0 \\ 0 & C_o \end{pmatrix}. \tag{2.23}$$

So the D_{ee} and D_{oo} of the even-odd literature (Ref. [47], Eq. (18); Ref. [45], Eq. (4)) *are* the site-diagonal part: $D_{ee} = C_e$ and $D_{oo} = C_o$ are the blocks of the same C that appears in $A' = (D_{\rm PV}C_{\rm PV}^{-1})^\dagger DC^{-1}$, restricted to the even and to the odd sites. Since C is the same $L_s \times L_s$ matrix on every site (Sec. II.B.3), C_e and C_o are also the same matrix, applied on different sites. The same holds for $D_{\rm PV}$ with $C_{{\rm PV},e}$, $C_{{\rm PV},o}$.

With these blocks the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and $S_{\rm PV}$ is that of $D_{\rm PV}$. The injection ι puts a field on the even sites onto the whole lattice, with zero on the odd sites, and the projection $\pi = \iota^\dagger$ takes the even sites:

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e, \qquad \iota\pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \begin{pmatrix} \psi_e \\ 0 \end{pmatrix}, \tag{2.24}$$

Two identities are used throughout, $S^{-1} = \pi D^{-1}\iota$ and $(S_{\rm PV}^\dagger)^{-1} = \pi(D_{\rm PV}^\dagger)^{-1}\iota$. To see them, solve $D\psi = \iota\eta_e = (\eta_e, 0)^{\mathsf T}$. The odd rows give $\psi_o = -C_o^{-1}D_{oe}\psi_e$, and the even rows then read $S\psi_e = \eta_e$. Hence $S^{-1}\eta_e = \pi D^{-1}\iota\eta_e$, and the same argument for $D_{\rm PV}^\dagger$ gives the second identity.

Two forms. The site-diagonal factor can stand on either side of S. This gives the right and the left form,

$$\tilde{A}_R = (S_{\rm PV}C_{{\rm PV},e}^{-1})^\dagger\,(S\,C_e^{-1}), \qquad \tilde{A}_L = (C_{{\rm PV},e}^{-1}S_{\rm PV})^\dagger\,(C_e^{-1}S). \tag{2.25}$$

$\tilde{A}_R$ is the even-odd counterpart of A'. $C_e^{-1}S$ is the usual left-preconditioned even-odd operator, which Boyle and Yamaguchi found worse than S itself for CG on the normal equation [40].

With Neff's α. Replace D by DQ. Since Q acts within a site and is diagonal in s, the blocks of DQ are C_eQ_e, $D_{eo}Q_o$, $D_{oe}Q_e$ and C_oQ_o. Its Schur complement is therefore

$$C_eQ_e - D_{eo}Q_o(C_oQ_o)^{-1}D_{oe}Q_e = SQ_e, \tag{2.26}$$

and its diagonal block is C_eQ_e. In the right form the two factors of Q_e cancel: $SQ_e(C_eQ_e)^{-1} = SC_e^{-1}$. So $\tilde{A}_R$ does not depend on α. In the left form they do not cancel. With $(C_eQ_e)^{-1}SQ_e = Q_e^{-1}C_e^{-1}SQ_e$ and Q Hermitian,

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e\,(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-2}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-1}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ only}. \end{cases} \tag{2.27}$$

For $D\psi = \eta$, with $\eta'_e = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger\eta'_e$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger C_e^{-1}\eta'_e$ (operators with Q when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or y (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

Choice of form. The two forms serve different purposes. The factorized correction of Sec. III.A is exact for the right form (Eq. (3.7)), but Q cancels in it, so the right form cannot use Neff's α. The left form keeps α. The correction is not exact for it, yet it needs about as many iterations as the right form, and with α it can save iterations. The outer solver therefore uses the left form in practice. The measurements of Sec. V are on the left form unless stated.

α in CG and in the multigrid. The left Schur factor of DQ is a similarity transform of that of D:

$$\mathcal{K}^{(\alpha)} = (C_eQ_e)^{-1}SQ_e = Q_e^{-1}\mathcal{K}Q_e, \qquad \mathcal{K} = C_e^{-1}S. \tag{2.28}$$

So $\mathcal{K}^{(\alpha)}$ and $\mathcal{K}$ have the same eigenvalues but different singular values, because Q_e is not unitary. CG on the normal equation converges at a rate set by the singular values and sees the change at every mass. In the multigrid the coarse correction removes the low modes for any α, so we expect α to help it less than CG. The measurements bear this out (Sec. V.B.1).

III. The two-grid method

A multigrid preconditioner removes the error of an iterate in two parts. A few steps of a cheap iteration on the fine lattice, the *smoother*, damp the high modes. The low modes, which make CG slow, are removed by solving a smaller problem on a coarse lattice. The fine lattice is cut into blocks, here of 4^4 sites with the whole fifth dimension; each block becomes one coarse site, whose degrees of freedom are a few approximate low modes restricted to the block, the *test vectors*. The prolongation $\mathcal{P}$ maps a coarse vector to the fine lattice through them, and the coarse operator of a fine operator X is its Galerkin projection $\mathcal{P}^\dagger X\mathcal{P}$. A few vectors per block suffice because low modes are locally coherent [20]. Finding them, the *setup*, is paid once per gauge field. The setup uses only the plain D and D_{PV}; Neff's α enters only the outer operator $\tilde{A}_L^{(\alpha)}$, the smoother that works on it, and the site-local factors of the correction (3.2), so one setup serves both forms and every α.

A. The factorized coarse-grid correction and the prolongation

1. Why factorize

Solving on the even sites halves the problem and improves the conditioning, and red-black CG, the solver to beat, works there. But the Schur complement reaches two hops, so the existing methods make a choice (Table 1). HDCG solves on the even sites and coarsens the squared Schur complement, with an 81-point coarse stencil [38]. The methods with a small coarse stencil work on all lattice sites: HDCR [39] coarsens $\gamma_5 RD$, Boyle and Oare [43] coarsen $D_{\mathrm{PV}}^\dagger D$, and MG-PV [41, 40] coarsens D and

TABLE 1. Multigrid methods for domain-wall fermions: the system solved, the operator that is coarsened, and its coarse stencil, for blocks that contain the whole fifth dimension (except MG-PV, whose coarse lattice keeps the fifth dimension) and are at least two sites wide in each direction (4^4 for HDCG). S is the Schur complement on the even sites.

method	system solved	operator coarsened	coarse stencil
HDCG [38, 36]	even sites, $S^\dagger S$	$S^\dagger S$ (four hops)	81 points
HDCR [39]	all sites, $\gamma_5 RD$	$\gamma_5 RD$, $c = 0$ (one hop)	9 points
MG-PV [41, 40]	all sites, $D_{\mathrm{PV}}^\dagger D$	D and D_{PV} apart, $c = 0$	9 points per slice
Boyle and Oare [43]	all sites, $D_{\mathrm{PV}}^\dagger D$	$D_{\mathrm{PV}}^\dagger D$ (two hops)	33 points
this work	even sites, $\tilde{A}_R$ or $\tilde{A}_L$	D and D_{PV} apart (one hop each)	9 points each

TABLE 2. Site-local factors of the correction (3.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1}\iota$; with α the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

D_{PV} separately on a coarse lattice that keeps the fifth dimension. We keep the separate coarse operators of D and D_{PV}, use them for the even-odd operator, and make the coarse lattice four-dimensional.

The point counts follow from the hops. An operator that reaches k hops on the fine lattice couples coarse sites whose coordinates differ by at most 1 in at most k directions. For $k = 1$ and 2 this gives $1 + 8 = 9$ and $1 + 8 + 24 = 33$ points, and for $k \geq 4$ all $3^4 = 81$.

The correction. By $S^{-1} = \pi D^{-1}\iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e\, S^{-1}\, (S_{\mathrm{PV}}^\dagger)^{-1}\, C_{\mathrm{PV},e}^\dagger = C_e\, \pi D^{-1}\iota\; \pi (D_{\mathrm{PV}}^\dagger)^{-1}\iota\; C_{\mathrm{PV},e}^\dagger, \tag{3.1}$$

a product of two inverses of one-hop operators and site-local factors. The Galerkin operator of $\tilde{A}_R$ itself would have a large coarse stencil, since $\tilde{A}_R$ reaches four hops. The Galerkin operators of D and D_{PV} are nearest-neighbor, since D is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any α, the correction of a residual r on the even sites is

$$e = O_L\; \pi\mathcal{P}\, \mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger\iota\;\; O_M\;\; \pi\mathcal{P}\left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}\right)^{-1}\mathcal{P}^\dagger\iota\;\; O_R\, r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}, \tag{3.2}$$

with the site-local factors O_R, O_M, O_L of Table 2. They come from writing each $\tilde{A}^{-1}$ as in Eq. (3.1). For example, for the left form with α in both operators, Eq. (2.27) gives

$$\tilde{A}_L^{(\alpha)\,-1} = Q_e^{-1}\, S^{-1}\, C_e Q_e^2 C_{\mathrm{PV},e}^\dagger\, (S_{\mathrm{PV}}^\dagger)^{-1}\, Q_e^{-1}, \tag{3.3}$$

so $O_R = Q_e^{-1}$, $O_M = C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$ and $O_L = Q_e^{-1}$. The two coarse inverses are those of the plain D and D_{PV}; within the correction, α is in these factors only. Figure 3 follows a residual through Eq. (3.2). It moves between the even sites, all lattice sites and the coarse lattice twice, and the result of the first coarse solve, taken on the even sites, is the right-hand side of the second.

2. Aggregation and prolongation

We divide the fine lattice into rectangular four-dimensional blocks. Each block contains the whole fifth dimension and becomes one coarse site. We split each test vector v into its Γ_5 eigencomponents

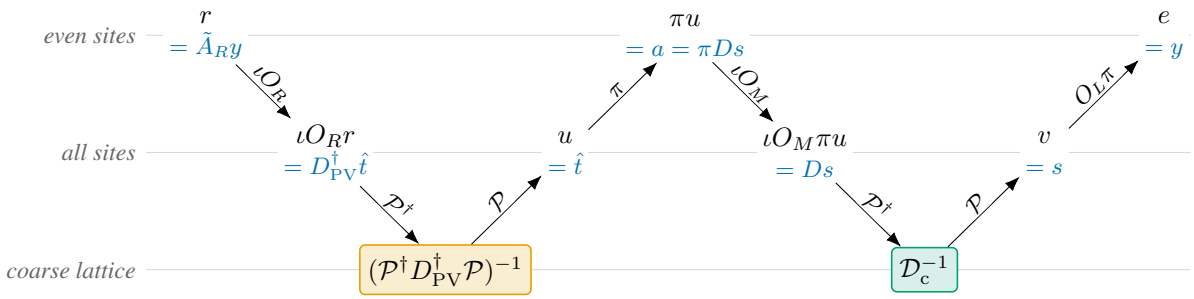


FIG. 3. One factorized coarse correction, from r (left) to e (right), which is Eq. (3.2) read from the right, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1}\mathcal{P}^\dagger \iota O_R r$ is the result of the first coarse solve, that of the adjoint Pauli–Villars operator, and its even part, with O_M, is the right-hand side of the second, that of the quark operator, whose result is $v = \mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger \iota O_M \pi u$; $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Eq. (3.7); $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

$v_\pm = \frac{1}{2}(1 \pm \Gamma_5)v$ and orthonormalize the $+$ parts and the $-$ parts separately within each block. With n_v test vectors a coarse site then carries $2n_v$ components: the first n_v of positive and the last n_v of negative Γ_5-chirality.

As in the chirality split of Wilson multigrid [28], each column of $\mathcal{P}$ is then an eigenvector of Γ_5. Since $\Gamma_5 = \gamma_5 R$, the condition $\Gamma_5 v_\pm = \pm v_\pm$ reads $v_\pm(L_s - 1 - s) = \pm\gamma_5 v_\pm(s)$. So each part is fixed by its values for $s < L_s/2$, and $\mathcal{P}$ is stored only there, at half the memory. On the even sites the transfer operators are $\mathcal{P}^\dagger\iota$ and $\pi\mathcal{P}$.

3. *The basis* $\{s, \hat{t}\}$

The prolongation is built from vectors y_j on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. III.C). Two facts shape how we use them.

- $\mathcal{P}$ and the coarse operators act on all lattice sites, so each y must be extended, or *lifted*, to all lattice sites. We fill the odd sites so that the fine operator maps the lifted vector to the even-odd quantity, just as ψ_o is rebuilt from ψ_e after an even-odd solve.
- The correction (3.2) contains two Galerkin solves, one with $D_{\mathrm{PV}}^\dagger$ and one with D. Each is exact only on vectors in $\operatorname{range}\mathcal{P}$. So we lift each y in two ways, one for each solve.

The two lifts are

$$s = \left(C_e^{-1} y,\ -C_o^{-1} D_{oe} C_e^{-1} y\right), \tag{3.4}$$

$$\hat{t} = \left(a,\ -(C_{\mathrm{PV},o}^\dagger)^{-1}(D_{\mathrm{PV},eo})^\dagger a\right), \qquad a = \pi D s. \tag{3.5}$$

The lift s is chosen so that Ds vanishes on the odd sites. With the blocks of Eq. (2.23),

$$Ds = \begin{pmatrix} C_e C_e^{-1} y - D_{eo} C_o^{-1} D_{oe} C_e^{-1} y \\ D_{oe} C_e^{-1} y - C_o C_o^{-1} D_{oe} C_e^{-1} y \end{pmatrix} = \begin{pmatrix} S C_e^{-1} y \\ 0 \end{pmatrix} = \iota\, S C_e^{-1} y. \tag{3.6}$$

In the same way $\hat{t}$ is chosen so that $D_{\mathrm{PV}}^\dagger \hat{t}$ vanishes on the odd sites. The block form of $D_{\mathrm{PV}}^\dagger$ has $(D_{\mathrm{PV},eo})^\dagger$ as its odd-even block and $C_{\mathrm{PV},o}^\dagger$ as its odd diagonal block, and the same computation gives $D_{\mathrm{PV}}^\dagger \hat{t} = \iota\, S_{\mathrm{PV}}^\dagger a$.

The first coarse solve of the correction returns $\hat{t}$ and the second returns s (Eq. (3.7) below). Since $\hat{t}$ is made from Ds, the two lifts follow the order of the solves. With the y alone in the basis, each factor of the correction would be an approximation, and the errors of the two would multiply. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

The two lifts differ in the fifth dimension: their odd parts are built with C_o^{-1} and with $(C_{\mathrm{PV},o}^\dagger)^{-1}$, which by Eq. (2.18) carry each chirality toward opposite ends of the fifth dimension. So a space built from the s_j alone does not contain the $\hat{t}_j$.

The right form. Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$, let $\mathcal{D}_\mathrm{c}$ and $\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P}$ be invertible, and let both coarse systems in Eq. (3.2) be solved exactly, with the exact adjoint $\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P} = (\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger$. Then the right-form correction returns the vector it is built from,

$$e(\tilde{A}_R y) = y. \tag{3.7}$$

For any invertible $\mathcal{P}^\dagger X \mathcal{P}$ the map $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$, and we follow $r = \tilde{A}_R y$ through the correction with the factors of the right form, $O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$. First, $O_R r = C_{\mathrm{PV},e}^\dagger \tilde{A}_R y = S_\mathrm{PV}^\dagger S C_e^{-1} y = S_\mathrm{PV}^\dagger a$ and $\iota S_\mathrm{PV}^\dagger a = D_\mathrm{PV}^\dagger \hat{t}$, so the first coarse solve returns $\hat{t}$, whose even part is $\pi\hat{t} = a = \pi D s$. The second solve then has the right-hand side $\mathcal{P}^\dagger \iota \pi D s = \mathcal{P}^\dagger D s$, because Ds vanishes on the odd sites, and returns s, whose even part is $\pi s = C_e^{-1} y$. Finally $O_L = C_e$ gives $e = y$.

In practice. Equation (3.7) holds only approximately: the stored $\mathcal{P}$ contains s and $\hat{t}$ only up to the rounding of its reduced-precision storage, the coarse solves stop at a loose tolerance, and the Pauli–Villars solve is regularized.

The left form. For the left form with α in both operators, the same argument needs the quark lift s' of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C_{\mathrm{PV},e}^{\dagger\,-1} Q_e^{-2} C_e^{-1} S Q_e y'$ in $\operatorname{range}\mathcal{P}$ (Table 2 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ we have $s' = s$. But $u = C_{\mathrm{PV},e}^{\dagger\,-1} C_e^{-1} a$ differs from a by a site-local matrix that mixes s and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors, and the correction is not exact for the left form. The left form converges nonetheless, at $\alpha = 1$ with the spectrum of the right form (Sec. V.D.2), and with the same vectors y it still needs $\hat{t}$ in the basis (Sec. V.C.2).

4. The coarse operators

We assemble $\mathcal{D}_\mathrm{c} = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{D}_\mathrm{c}^\mathrm{PV} = \mathcal{P}^\dagger D_\mathrm{PV} \mathcal{P}$ from the hops of the fine operator between neighboring blocks and from its part within a block. Each coarse site has a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. The adjoint factor $\mathcal{P}^\dagger D_\mathrm{PV}^\dagger \mathcal{P} = (\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger$ is the conjugate transpose of these blocks. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity, and both coarse operators have an even-odd form of their own.

B. The coarse solvers

We solve both coarse systems only approximately, in reduced precision, to a loose relative residual or a maximum number of iterations. Reaching the tolerance is not required. The two are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor. The default, used for all results unless stated, is BiCGStab on the even-odd form of the quark factor and CG on the regularized normal equation (3.9) of the Pauli–Villars factor, both to $|r|^2/|b|^2 < 10^{-2}$ (Algorithms 7 and 6, Appendix D).

1. The quark factor

Write $\mathcal{D}_\mathrm{c} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_\mathrm{c} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe} c_e), \tag{3.8}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks A_{ee}^{-1}. BiCGStab [22] is Algorithm 7 of Appendix D.

2. The Pauli–Villars factor

The system is $Xw = g$ with $X = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^{\dagger}$. We solve the regularized normal equation

$$\left(X^{\dagger}X + \tau\right) w = X^{\dagger} g \tag{3.9}$$

by CG on all coarse sites, with $\tau = 10^{-3}$. Against the exact solve, the shift multiplies the component along each right singular vector of X by $\sigma^2/(\sigma^2 + \tau)$, so it damps most the directions with σ below about $\sqrt{\tau}$. The Galerkin factor can have far smaller singular values than the fine D_{PV} (Sec. V.C.3).

3. Levels below the coarse lattice

The quark factor may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^{\dagger} s_j$ and $\mathcal{P}^{\dagger} \hat{t}_j$, split by chirality and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With Q_{ℓ} the prolongation into level $\ell - 1$ and B_{ℓ} the diagonal blocks of level ℓ, we use a V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^{\dagger} r, \qquad M_{\ell} r = z + B_{\ell}^{-1}(r - \mathcal{D}_{\ell} z); \tag{3.10}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system.

Levels instead of deflation. The levels below compress the low modes further by aggregation and invert the last level exactly, by LU in double precision, so every mode that reaches that level is treated without computing any eigenvector. The alternative, deflating the coarse factors with a few of their eigenvectors, we have not tried with the basis $\{s, \hat{t}\}$. Boyle and Oare [43] use a large coarsest level that is inverted densely in the role of deflation.

C. Setup

The setup produces the vectors $y_1, \ldots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$ and the coarse operators. It always uses the right form; the same setup serves the left forms and every α (Sec. III.A). It has two stages. A Chebyshev filter finds approximate low modes without any multigrid. A bootstrap then improves them with the cycle built from them, so that they also cover what the first coarse space misses.

1. Chebyshev-filtered subspace iteration

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, that is the low eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^{\dagger} \tilde{A}_R$. We find them by filtering. Let T_d be the Chebyshev polynomial of degree d and $\ell(x) = \big(2x - (\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})\big)/(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$. Then $|T_d(\ell(x))| \le 1$ on $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$, while it grows quickly below λ_{lo}. So $T_d(\ell(H_n))$ amplifies the low modes against the rest. We apply it by the three-term recurrence.

Algorithm 1. LOWMODES: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

$X \leftarrow n$ random vectors on the even sites
$\lambda_{\mathrm{hi}} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration); $\lambda_{\mathrm{lo}} \leftarrow$ a small fraction of λ_{hi}
for each round **do**
$d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}}$, within fixed bounds
$X \leftarrow T_d(\ell(H_n))\, X$; orthonormalize X through its Gram matrix
$G \leftarrow X^\dagger H_n X = V \Theta V^\dagger$; $X \leftarrow XV$ ▷ Rayleigh–Ritz
$\lambda_{\mathrm{lo}} \leftarrow \theta_{\max}$ ▷ largest Ritz value
end for
return X, Θ

2. *Bootstrap*

Each bootstrap round has three steps. It builds the basis and the two-grid cycle of Sec. III.D from the current y. It replaces every y_j by a few FGMRES steps (Algorithm 5) on $\tilde{A}_R x = y_j$ from $x = 0$, preconditioned by that cycle. And it orthonormalizes the new y through their Gram matrix ($y \leftarrow yV\Lambda^{-1/2}$ with $y^\dagger y = V\Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

Algorithm 2. Setup of the two-grid method.

$y \leftarrow$ LOWMODES(H_n)
for each bootstrap round **do**
build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of y and the cycle $\mathcal{M}$ on it
for $j = 1, \ldots, n$ **do**
$y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
end for
orthonormalize y
end for
compute $s_j, \hat{t}_j$ by Eqs. (3.4)–(3.5); split by Γ_5; orthonormalize per block $\rightarrow \mathcal{P}$
assemble $\mathcal{D}_{\mathrm{c}}$, $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$, the inverses of their diagonal blocks, their even-odd forms
optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$

D. The Chebyshev smoother and the cycle

1. *Smoother*

The smoother is ν steps of Chebyshev iteration [48, 49] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$: λ_{hi} slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/\kappa_s$ for a chosen ratio κ_s. With $\theta = \frac{1}{2}(\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}})$, $\delta = \frac{1}{2}(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step k is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A} d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta}\, r_{k+1}, \tag{3.11}$$

The last step only updates x, so ν steps cost $\nu - 1$ applications of $\tilde{A}$. The recurrence keeps four fine vectors, x, r, d and $\tilde{A}d$, whatever ν, and needs no inner products; the smoother is the same linear map at every application. A Krylov smoother, such as the fixed number of GCR steps of the Pauli–Villars multigrid of Ref. [40], keeps two more fine vectors per step, which in a fixed amount of memory lowers the largest volume that can be solved. We use only a real focal interval. The ratio κ_s is chosen by hand. The polynomial damps the part of the spectrum above λ_{lo} and leaves the rest to the coarse correction, and an interval that is too wide or too narrow can make the outer solve stagnate. The interval that works differs between the two setups of Sec. V (Secs. V.A and V.C.4).

2. *Cycle*

Algorithm 3. Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

1: $y \leftarrow p(\tilde{A})\, b$ ▷ pre-smoothing
2: $r \leftarrow b - \tilde{A}y$
3: $y \leftarrow y + e(r)$ ▷ correction (3.2)
4: $r \leftarrow b - \tilde{A}y$
5: $y \leftarrow y + p(\tilde{A})\, r$ ▷ post-smoothing

One cycle costs 2ν applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

E. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as a right preconditioner, and we rebuild ψ as in Sec. II.C. The cycle $\mathcal{M}$ changes from one application to the next, so the outer method must be flexible: FGMRES or flexible GCR (Algorithms 5 and 4, Appendix D), which keep the preconditioned directions. The method allows any number k of kept directions; the largest k that fits in memory is the natural choice. Each kept direction costs two fine vectors, about 1 GB on $24^3 \times 48$, where the multigrid with GCR(2) already needs 20–23 of the 24 GB of device memory, so $k = 2$ there. For solves with D we use $k = 2$ on every volume, so that the volumes compare directly. A larger k changes only the outer solver and leaves the cycle as it is; on 12^4 it changes the median iterations by at most one (Sec. V.D.4).

Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [44]. Otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [35, 50], and a truncated method can stagnate [51]. We monitor convergence on the even-odd system and evaluate the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, with the plain D after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE). It is robust but squares the condition number. It runs in double precision, or in mixed precision: CG in single precision with reliable updates in double precision. Appendix D compares it with CG on $S^\dagger S$.

IV. Solving with $D^\dagger$

HMC needs $(D^\dagger D)^{-1}$ at every force evaluation, so it solves with $D^\dagger$ as often as with D. The multigrid of D also preconditions solves with $D^\dagger$, with no second setup, coarse operator or cycle. The construction needs only an approximate inverse of D, so it applies to any preconditioner built for the unsquared operator. The tool is the exact relation (2.14) between D and $D^\dagger$.

A. The idea

For Wilson fermions $D_W^\dagger = \gamma_5 D_W \gamma_5$, so $D_W^{-\dagger} = \gamma_5 D_W^{-1} \gamma_5$, and a solver for D_W solves with $D_W^\dagger$ as well. For Möbius fermions this fails at non-zero c (Sec. II.B.2). We use $\Gamma_5 D^{-1} \Gamma_5$ anyway, as a

preconditioner for $D^\dagger$, and Eq. (4.2) below says how good it is.

Multiply $D^\dagger$ by $\Gamma_5 D^{-1}\Gamma_5$ and use $D^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger$ from Eq. (2.14) and $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$D^\dagger\, \Gamma_5 D^{-1}\Gamma_5 = \Gamma_5 D_-^{-1} D\, D_- D^{-1}\, \Gamma_5 = \Gamma_5 \big(1 + D_-^{-1}[D, D_-] D^{-1}\big)\Gamma_5. \tag{4.1}$$

The commutator follows from $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$: $[D, D_-] = -D_-[Y, D_-] = -c\, D_-[Y, W]$. The factor D_- cancels, and

$$\boxed{D^\dagger\, \Gamma_5 D^{-1}\Gamma_5 = \Gamma_5\, (1 - N)\, \Gamma_5, \qquad N = c\, [Y, W]\, D^{-1}.} \tag{4.2}$$

Equation (4.2) has three properties used below:

- D_-^{-1} has dropped out. We never invert D_-, which we cannot do (Sec. IV.C).
- $[Y, W] = -\frac{1}{2}[Y, K]$ is a bounded operator: one hop in four dimensions and one in the fifth. So N is large only where D^{-1} is large, on the low modes of D.
- On a low mode v the commutator acts through the γ_μ in K, which flip chirality. A low mode of D carries on each wall the chirality bound there, P_- at $s = 0$ and P_+ at $s = L_s - 1$; $[Y, W]v$ carries on each wall the opposite chirality, which no low mode has there. So N takes the low modes out of their span, and N^2 is small. Then $(1 - N)^{-1} \approx 1 + N$, and a Krylov method needs only a few steps more than for the identity.

The third remark is a heuristic; Sec. V.B.2 tests it and finds that it fails on 12^4 in setup S at $m = 0.001$.

B. On the even sites

The solves of HMC are on the even sites, so we need the same idea for the Schur complement S. The blocks of $D^\dagger$ are $C_e^\dagger$, $C_o^\dagger$ on the diagonal and $D_{oe}^\dagger$, $D_{eo}^\dagger$ off it, so its Schur complement is $S^\dagger$, and $S^{-1} = \pi D^{-1} \iota$. Because Γ_5 acts within a site, it commutes with π and ι, and

$$S^{-\dagger} = \pi D^{-\dagger} \iota \approx \pi\, \Gamma_5 D^{-1}\Gamma_5\, \iota = \Gamma_5 S^{-1}\Gamma_5. \tag{4.3}$$

The solve of $D^\dagger w = r$ then goes as follows.

1. Form $r'_e = r_e - D_{oe}^\dagger (C_o^\dagger)^{-1} r_o$, so that $S^\dagger w_e = r'_e$.
2. Write $w_e = (C_e^\dagger)^{-1} z$. Then $\mathcal{K}^\dagger z = r'_e$ with $\mathcal{K} = C_e^{-1} S$, the operator HMC already uses.
3. Solve for z by flexible GCR with the preconditioner $C_e^\dagger\, \Gamma_5 S^{-1}\Gamma_5$. The inverse S^{-1} is one application of the cycle $\mathcal{M}$ of $\tilde{A}_R$: $S^{-1} t = C_e^{-1} \tilde{A}_R^{-1} (S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t \approx C_e^{-1} \mathcal{M}\, (S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t$.
4. Recover $w_e = (C_e^\dagger)^{-1} z$ and $w_o = (C_o^\dagger)^{-1}(r_o - D_{eo}^\dagger w_e)$.

Each GCR step costs one cycle, as a step of the solve with D does.

C. Why D_- cannot be inverted

Eq. (2.14) seems to offer more: the exact null vectors of $D^\dagger$, and an exact reduction of $D^\dagger w = r$ to a solve with D. Both need D_-^{-1}, and D_- cannot be inverted in practice because of its spectrum. Since $D_- = cW - 1 = c\,(W - 1/c)$, D_- is c times the Wilson operator of mass $m_0 = -M_5 - 1/c$: -3 in setup S and -3.8 in setup B. The free eigenvalues of that operator,

$$m_0 + \sum_\mu (1 - \cos p_\mu) \pm i \Big(\sum_\mu \sin^2 p_\mu \Big)^{1/2}, \tag{4.4}$$

have real parts from m_0 to $m_0 + 8$, here from -3 to 5. Near real part zero they have imaginary parts of both signs, so the spectrum surrounds the origin. By the maximum principle no polynomial p with $p(0) = 1$ is smaller than 1 everywhere on a closed curve around the origin, so a Krylov method reduces the residual only by resolving single eigenvalues, with a degree of the order of their number (Sec. II.A). The preconditioner of Eq. (4.2) avoids D_-^{-1} altogether.

D. New!!! $D^\dagger$ through D itself

We found a second way to solve with $D^\dagger$ from D itself without a second setup, and this one is exact. Write the operator of Eq. (2.8) as

$$D = 1 + Y + W B_5, \qquad B_5 = b - cY. \tag{4.5}$$

With Eq. (2.6),

$$(B_5\psi)_s = b\,\psi_s + c\,\big(P_-\psi_{s+1} + P_+\psi_{s-1}\big), \tag{4.6}$$

where the terms that wrap around carry $-mc$ instead of c. For $L_s = 4$, as a matrix in s (compare Eq. (2.7)),

$$B_5 = \begin{pmatrix} b & cP_- & 0 & -mcP_+ \\ cP_+ & b & cP_- & 0 \\ 0 & cP_+ & b & cP_- \\ -mcP_- & 0 & cP_+ & b \end{pmatrix}. \tag{4.7}$$

On $\psi^+ = P_+\psi$ it is lower bidiagonal with one corner element, on ψ^- upper bidiagonal, as C in Eq. (2.15). B_5 acts within each four-dimensional site, on the fifth dimension and the spin. The properties of Sec. II.B.2 give $\Gamma_5 Y^\dagger \Gamma_5 = Y$ and $\Gamma_5 W^\dagger \Gamma_5 = W$, hence $\Gamma_5 B_5^\dagger \Gamma_5 = B_5$ and $\Gamma_5 D^\dagger \Gamma_5 = 1 + Y + B_5 W$. Because B_5 is a polynomial in Y, it commutes with $1 + Y$, so $B_5 D B_5^{-1} = 1 + Y + B_5 W$ as well, and

$$\boxed{\Gamma_5 D^\dagger \Gamma_5 = B_5\, D\, B_5^{-1}, \qquad (D^\dagger)^{-1} = \Gamma_5 B_5\, D^{-1} B_5^{-1}\, \Gamma_5.} \tag{4.8}$$

The solve of $D^\dagger w = r$ is then the solve of $Dx = B_5^{-1}\Gamma_5 r$, and $w = \Gamma_5 B_5 x$. Compared with Eq. (2.14), the factor between D and $D^\dagger$ is B_5 instead of $D_- = cW - 1$. D_- couples neighboring sites and cannot be inverted (Sec. IV.C); B_5 has the form $d_+ + d_- Y$ of the site-local part C of Sec. II.B.3, with $d_+ = b$ and $d_- = -c$, so the exact inverse of C inverts it, and by the argument of Eq. (D.1)

$$\kappa(B_5) \le \frac{b+c}{b-c} = 2 \qquad (b = 1.5,\ c = 0.5). \tag{4.9}$$

The bound holds for $b > c$. For $b = c$ (Boriçi's operator) it is not finite, but Eq. (4.8) still holds: B_5 is singular only if b/c is an eigenvalue of Y, and the eigenvalues of Y have modulus $m^{1/L_s} < 1$, so $B_5 = b(1 - Y)$ is invertible for $m < 1$. Its condition number grows with L_s; numerically $\kappa(B_5) \approx 11$ for $L_s = 8$ and 21 for $L_s = 16$ at $m = 0.001$, against 1.9 and 2.0 for $(b, c) = (1.5, 0.5)$. Equation (4.8) needs b and c independent of s, as they are here; it does not hold for the s-dependent coefficients of Zolotarev domain-wall fermions.

On the even sites the relation carries over directly. Γ_5 and B_5 act within a site, so they commute with the restriction π and the embedding ι, and

$$S^{-\dagger} = \pi D^{-\dagger} \iota = \Gamma_5 B_5\, S^{-1} B_5^{-1}\, \Gamma_5 \tag{4.10}$$

with B_5 on the even sites. A source with nothing on the odd sites keeps that property under $B_5^{-1}\Gamma_5$, so the solve with $D^\dagger$ of HMC is one even-site solve with D, with the same cycle $\mathcal{M}$ and the same Krylov method, and it costs what a solve with D costs. The preconditioner $\Gamma_5 \mathcal{M} \Gamma_5$ of Eq. (4.2) and its analysis are then no longer needed for the solver.

Table 3 compares the two ways on the GPU, with GCR(2), one cycle per step and $|r|^2/|b|^2 < 10^{-10}$, on the right-hand sides of Sec. V.B.2. The solve through D needs the iterations of a solve with D, within two either way, because it is one. The preconditioner $\Gamma_5 \mathcal{M} \Gamma_5$ needs two to four more on the random right-hand side and one to five more on the residual, and on 12^4 in setup S it stalls on the residual of all four configurations, which the solve through D does in 7 to 11 iterations. In time the solve through D takes 0.5–0.8 of the solve with $\Gamma_5 \mathcal{M} \Gamma_5$ on every lattice and right-hand side (medians, e.g. 5.2 against

TABLE 3. GCR(2) iterations of solves with D and with $D^\dagger$ from one setup of D on the GPU, one cycle per step, to $|r|^2/|b|^2 < 10^{-10}$, medians over four configurations (in brackets the number that do not converge in 300 iterations; "stall": all four). "rand": random right-hand side; "res": the residual of one sweep of $D^\dagger D$ (Sec. V.B.2). 16^4 at $m = 0.01$, the others at $m = 0.001$; 24^4 and $24^3 \times 48$ with the levels below the coarse lattice.

		D		$D^\dagger$ by $\Gamma_5\mathcal{M}\Gamma_5$		$D^\dagger$ through D, Eq. (4.8)	
lattice	setup	rand	res	rand	res	rand	res
12^4	S	5	7	11 [1]	stall	5	8
	B	7	9	11	14	7	10
16^4	S	5	6.5	8.5	10.5	5	7.5
	B	7	8.5	10	12	7	10
24^4	S	7.5	12.5	10.5	14	7.5	11
	B	11	17.5	14.5	20	11	16.5
$24^3 \times 48$	S	11	19	13.5	20.5	10	17
	B	11	18	14.5	21	11	17

TABLE 4. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: quenched, plaquette action at $\beta = 6.0$, on 12^4, 16^4, 24^4 and $24^3 \times 48$, four each; their plaquettes, t_0 and topological charges are in Appendix C.

	setup S (stout)	setup B (bare)
M_5	1.0	1.8
gauge smearing	stout, $\rho = 0.1$, 3 steps	none
Chebyshev smoother ν, κ_s	8, 10	12, 30
multigrid (both setups)		
aggregates	4^4 ($24^3 \times 48$: levels below with super-blocks 2, 3)	
test vectors	32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$)	
setup	filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps	
coarse solves	single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$; Pauli–Villars $\tau = 10^{-3}$	
precision	$\mathcal{P}$ half, cycle single, outer solver double	
outer solver	GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$	

7.7 s and 8.8 against 11.1 s on $24^3 \times 48$ in setup S), since each of its steps is a step of the solve with D. In setup S on 24^4 and $24^3 \times 48$ the solve $Dx_0 = w_0$ of the sweep does not converge in 2000 iterations on two of the four configurations, the stall of D on right-hand sides $D^{-\dagger}b$; the residual right-hand side of those two is that of the unconverged sweep.

V. Numerical results

A. Setups

Numerical setup. The quenched configurations use the plaquette action at $\beta = 6.0$, a lattice spacing of about 0.1 fm; Table 7 lists the configurations. Results are means, medians or ranges over the four configurations of each volume as the captions state; the solvers without the multigrid and the pseudospectra use configuration 1. The two setups of the Möbius operator (Table 4) share $(b, c) = (1.5, 0.5)$, the choice of the JLQCD and the RBC/UKQCD Collaborations, and $L_s = 8$.

- *Setup B (bare links)* uses $M_5 = 1.8$, the value typical of simulations on unsmeared links [53].

TABLE 5. Point-source propagators (12 columns) on $24^3 \times 48$: the multigrid (setup seconds, iterations and seconds per column) against the mixed-precision CGNE (iterations and seconds per column), the wall time of the whole propagator, and the largest relative difference of the pseudoscalar correlator at $t = 0 \dots 3$. At $m = 0.001$: ranges over the four configurations and the 12 columns; at $m = 0.01$: configuration 1. Both stop at $|r|^2/|b|^2 < 10^{-18}$ of their own systems (for CGNE, its normal equations).

		MG			CGNE, mixed		wall (s)		
	m	setup	it	s	it	s	MG	CGNE	$\lvert\Delta C\rvert/C$
S	0.001	203–222	22–48	11.1–26.4	3634–6400	52.1–91.6	382–505	678–1048	$\leq 1 \times 10^{-6}$
B	0.001	204–207	22–30	10.7–14.5	4354–5574	62.4–79.8	363–398	803–973	$\leq 7 \times 10^{-7}$
S	0.01	194	14	5.6–5.8	1337–1388	19.8–20.5	283	264	2×10^{-7}
B	0.01	204	17	7.8–7.9	1641–1745	24.2–25.7	318	324	4×10^{-8}

- *Setup S (stout links)* smears the links of the fermion operator (stout, $\rho = 0.1$, three steps, Sec. II.B.5) and uses $M_5 = 1.0$, as in the JLQCD ensembles of Ref. [5].

Both suppress the near-zero modes of $W = D_W(-M_5)$, by moving M_5 (B) or by smoothing the field (S), and so give the multigrid low modes of different character; they need different smoother intervals (Table 4).[1] Timings are from one NVIDIA data-center GPU with 24 GB of device memory; a second GPU was used for iteration counts and spectral quantities only. Every setup uses the same random seed except where other seeds are stated (Secs. V.C.3 and V.C.4).

B. Performance

1. Comparison with CGNE

On $24^3 \times 48$ the multigrid is 4.1–7.3 times faster than mixed-precision CGNE at $m = 0.001$, and its setup is recovered within one point-source propagator for $m \leq 0.005$.

The reference solver. It is CG on $\mathcal{K}^\dagger\mathcal{K}$, $\mathcal{K} = C_e^{-1}S$, the even-odd operator of Bridge++, with Neff's α as for the multigrid, in mixed precision. Boyle and Yamaguchi [40] used CG on $S^\dagger S$, which here needs more iterations (Appendix D).

Mass dependence. The multigrid removes most of the critical slowing down of CGNE (Fig. 4, Table 8). From $m = 0.1$ to $m = 0.001$ the iterations of CGNE grow by 20–32, those of the multigrid by 2.8–5.4. Its setup is paid back after 2.6–4.8 right-hand sides at $m = 0.001$. The multigrid is faster at every mass. The multigrid needs three times the device memory of CGNE.

A point-source propagator. At $m = 0.001$ the multigrid takes 0.40–0.56 of the wall time of CGNE for a whole propagator, setup included (Table 5). At $m = 0.01$ the two take about the same time for the 12 columns, setup included. The pseudoscalar correlators of the two solvers agree to 1×10^{-6} or better.

Volume. The advantage of the multigrid grows with the volume (Fig. 5, Table 9). In solve time, setup excluded, it is slower than CGNE (S) or level with it (B) on 12^4, faster on 16^4 (in setup S the ranges overlap), and clearly faster from 24^4 on. From 12^4 to $24^3 \times 48$ its iterations about double, while those of CGNE grow 7–25 times. Its setup grows with the volume. Neff's $\alpha = 0.6$ saves CG a larger share of its iterations (Fig. 10) than the multigrid (Table 8), as Sec. II.C expects.

[1] New!!! The multigrid also works for Boriçi's operator, $(b, c) = (1, 1)$, with a smoother interval chosen for it: in setup S a narrower one ($\kappa_s = 3$) is needed and then gives the iteration count of the Möbius operator, and in setup B the solve takes about four times the Möbius iterations. Its residual mass is not the reason; at $L_s = 8$ it is of the same size as the Möbius one in setup S and smaller in setup B.

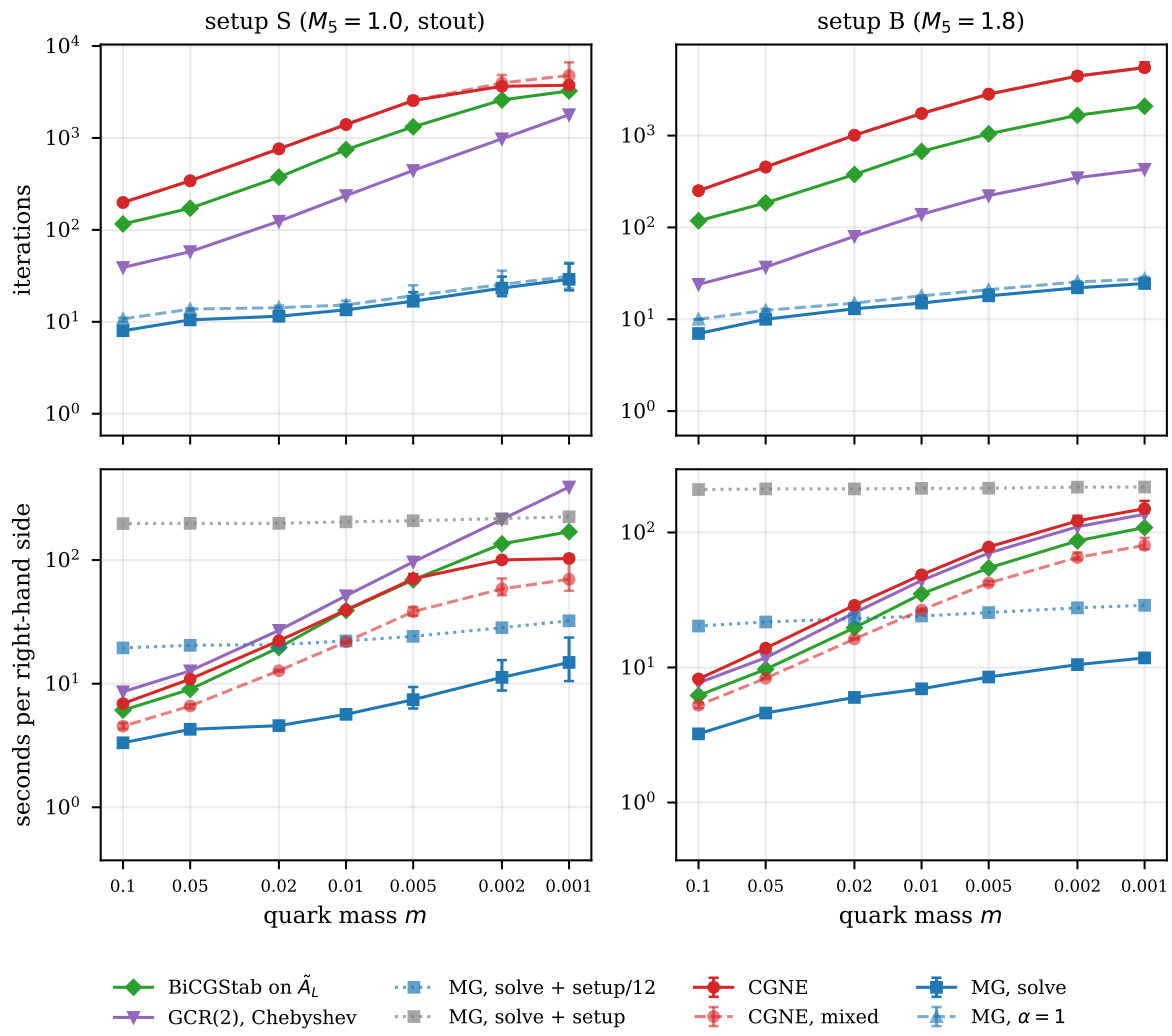


FIG. 4. The mass scan of Table 8, the mixed-precision CGNE, and the Krylov solvers on $\tilde{A}_L$ without the multigrid (Sec. V.C.1): iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$. CGNE and multigrid: means over the four configurations, bars from the smallest to the largest value; the solvers without the multigrid: configuration 1. "Solve + setup/12": the setup spread over the 12 columns of one point-source propagator.

2. *Solves with* $D^\dagger$

Single solves. One setup of D also solves with $D^\dagger$: flexible GCR on $D^\dagger$ with $\Gamma_5 \mathcal{M} \Gamma_5$, Eq. (4.2), one cycle per step, needs one to four iterations more than the solve with D on 16^4 at $m = 0.01$ (Table 10). The right-hand sides are a random vector b and the residual $r = b - D^\dagger D x_0$ of one sweep of the normal equations, where $D^\dagger w_0 = b$ and $D x_0 = w_0$ are solved to $|r|^2/|b|^2 < 10^{-10}$ (the first by GCR(8) with the cycle that has D and D_{PV} exchanged, Sec. VII), which leaves $|r|^2/|b|^2 = 1.4 \times 10^{-10}$–$3 \times 10^{-9}$; HMC meets right-hand sides of the second kind. On 16^4 on the random right-hand side more inner cycles per step do not reduce the count, which suggests that there the extra steps come from N; on the residual right-hand side two cycles per step save a few iterations. On 12^4 in setup S at $m = 0.001$ the residual right-hand side is harder. GCR(2) with one cycle per step does not converge, GCR(8) converges on three of the four configurations and GCR(2) with two cycles per step on two, and GCR(8) with two cycles per step needs 10 iterations on all four.

The exact map of Eq. (2.14) would give the null vectors of $D^\dagger$ and reduce $D^\dagger w = r$ to a solve with D, but it needs D_-^{-1}, which does not converge (Sec. IV.C).

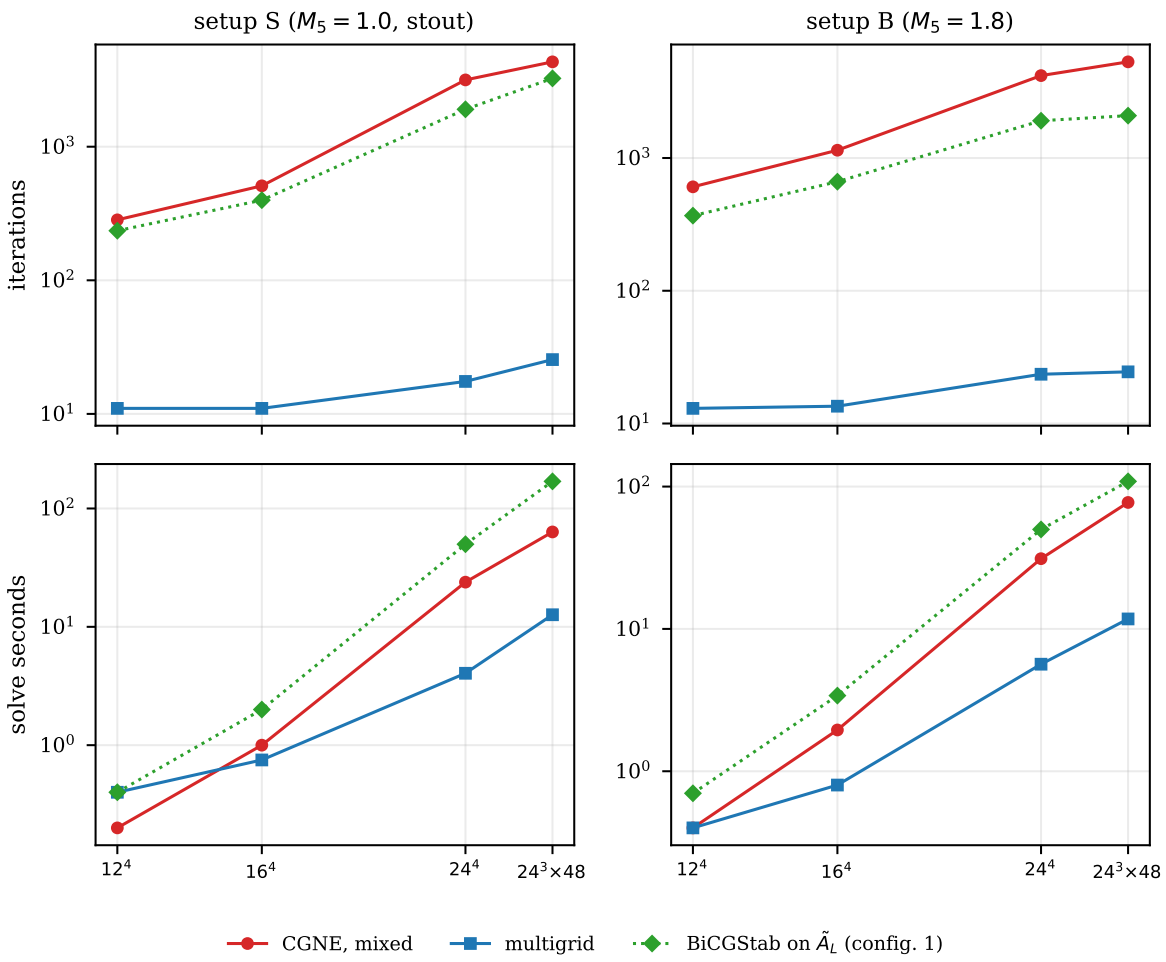


FIG. 5. Iterations (top) and solve seconds (bottom) at $m = 0.001$ against the volume, one random right-hand side. Mixed-precision CGNE and multigrid: median over the four configurations; BiCGStab on $\tilde{A}_L$: configuration 1. The ranges are in Table 9.

C. What each part contributes

The speed-up of Sec. V.B.1 rests on four choices beyond the operator A': the coarse correction, the lift $\hat{t}$ in the basis, the regularized Pauli–Villars solve, and the setup parameters.

1. Krylov on $\tilde{A}_L$ without the multigrid

The speed-up comes from the coarse correction. Without it a Krylov method on $\tilde{A}_L$ does not beat mixed-precision CGNE (Fig. 4). On $24^3 \times 48$ BiCGStab on $\tilde{A}_L$ alone needs fewer iterations than double-precision CGNE, about 0.4 of them, but each of its iterations applies $\tilde{A}_L$ twice, and it is slower than mixed-precision CGNE at every mass. GCR(2) with the Chebyshev smoother as its preconditioner needs fewer iterations than BiCGStab but more time.

2. Choice of the prolongation basis

The lift $\hat{t}$ lets a given number of vectors y suffice (Table 11). The exactness (3.7) needs both lifts s and $\hat{t}$ of each y in $\mathrm{range}\,\mathcal{P}$, but the left form used in practice is not exact (Sec. III.A.3). To see what $\hat{t}$ does there, we replace it by Ds, which vanishes on the odd sites, or drop it (Table 11). With $\{s\}$ alone $\mathcal{P}$ has half the columns; to compare at the same size we also take 32 vectors y, which doubles the vectors of the filter and of the bootstrap.

With the same vectors y, the two lifts give the fewest iterations on every volume. In setup S, replacing

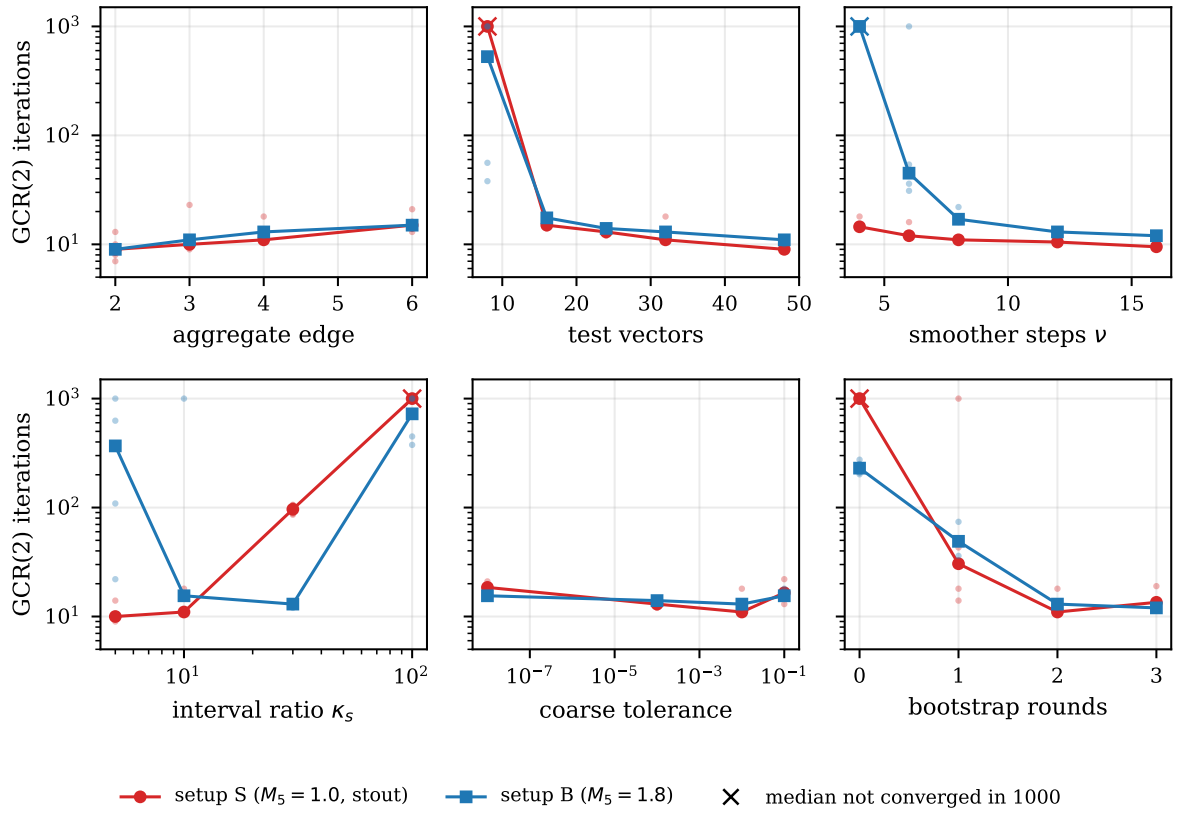


FIG. 6. GCR(2) iterations on $\tilde{A}_L$ when one parameter of Table 4 is changed, 12^4, $m = 0.001$, one random right-hand side. Lines: median over the four configurations; dots: the four configurations; a cross marks a median that does not converge in 1000 iterations. The default is the point of each line at 11 (S) and 13 (B) iterations. Values and setup and solve times in Table 15.

$\hat{t}$ by Ds or dropping it stagnates or slows down several times on one or two of the four 12^4 and 16^4 configurations and takes 2.5–3.2 times the iterations on $24^3 \times 48$; in setup B it costs more iterations. Twice as many vectors y with $\{s\}$ alone, the same number of columns, do as well on 12^4 and 16^4 and need fewer iterations on $24^3 \times 48$, but the filter and the bootstrap then handle twice the vectors and the setup takes 1.6–1.7 times as long.

3. *The regularized Pauli–Villars solve*

The regularized solve removes the slow solves (Table 13). Without the shift, with BiCGStab on the exact adjoint, setup S needs 84 to more than 2000 iterations on 12^4 and 29–50 on 16^4; with the shift it needs 10–18. Without smearing it needs 13–15 either way. The slow solves come with stout smearing, which gives the Galerkin factor a smallest singular value of $1/120$ to $1/41$ of that of the fine D_{PV} on 12^4 and $1/46$ to $1/9$ on 16^4; without smearing it is about 0.6 of it (Table 14). Among the smeared configurations this value does not order the iteration counts. The exact coarse solve amplifies the directions of small singular values, and the shift damps them. With the shift, 40 setups of setup S on 12^4 (ten seeds on each of the four configurations) need 10–17 iterations. Other values of τ from 10^{-4} to 3×10^{-2} give 11–19 iterations, except one run at $\tau = 10^{-2}$ (Table 12).

4. *Sensitivity to the parameters*

The method is robust within a window of each parameter and fails outside it (Fig. 6). The smoother interval must suit the setup, the bootstrap is necessary, and more accurate coarse solves do not help.

Fewer iterations are not always less time (Table 15). Smaller aggregates and more test vectors save iterations, but the larger coarse space makes the setup and each iteration longer.

The number of test vectors that suffices grows from 12^4 to 24^4 (Table 15 and Table 20); in setup B it does not grow further to $24^3 \times 48$, as local coherence of the low modes leads one to expect once the box is large enough. In setup B each count gives about the same iterations on both volumes. Setup S needs more vectors; with fewer than 32 it is slow or stagnates on both volumes.

Outside the window the failures can come from the outer solver alone. With $\kappa_s = 10$ in setup B, GCR(2) reaches a plateau on one of the four 12^4 configurations, and other setup seeds converge (Sec. V.D.4). This plateau comes with the regularization: without the shift the same setup converges.

D. Why the method converges

We infer that the spectrum and the pseudospectra of the preconditioned operator govern the convergence, since its field of values reaches into the left half-plane and the bound of Eisenstat, Elman and Schultz [44] for GCR does not apply. The quantities of this section are for $\alpha = 1$. $\tilde{A}_R$ does not depend on α, but $\tilde{A}_L$ at $\alpha = 0.6$ is a different operator (Eq. (2.27)) that the analysis does not cover; at $\alpha = 1$ the solves of Table 8 need 1–4 more iterations than at $\alpha = 0.6$ in the mean over the configurations. As usual for multigrid preconditioners of lattice Dirac operators, the convergence of the truncated flexible outer method is established numerically.

The *field of values* $W(X) = \{z^\dagger X z / z^\dagger z\}$ contains the spectrum, and its leftmost point is the smallest eigenvalue of the Hermitian part $(X + X^\dagger)/2$; GCR converges at every restart length when W lies in the right half-plane [44]. The *ε-pseudospectrum* is the set of z where $\sigma_{\min}(z - X) < \varepsilon$; for a normal X it is the ε-neighborhood of the spectrum, for a non-normal one it can be much larger, and it bounds full GMRES [50]. The condition number of an eigenvalue, $\kappa = 1/|y^\dagger x|$ for its normalized right and left eigenvectors, is 1 for a normal matrix and widens the pseudospectrum around the eigenvalue.

For these quantities $\mathcal{M}$ is a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision, the Pauli–Villars factor regularized as in Eq. (3.9), and $\mathcal{M}^\dagger$ built as its adjoint, so that $\langle z|\tilde{A}_R \mathcal{M} w\rangle$ and $\langle \mathcal{M}^\dagger \tilde{A}_R^\dagger z|w\rangle$ agree to about 10^{-10}.

1. Field of values

The coarse correction pushes the field of values into the left half-plane (Table 16). The field of values of $\tilde{A}_R$ itself reaches only just below zero, a little further on larger volumes, and the smoother is harmless, $\|I - \tilde{A}_R \mathcal{S}\| \approx 1$. Without the shift the coarse correction E is the identity on the vectors it is built from (Eq. (3.7)). With stout smearing it amplifies other directions, like a strongly oblique projection, by three orders of magnitude, without smearing by less than a hundred (both with the shift). The post-smoothing acts as a filter and brings the whole cycle back to within a factor of ten of the identity. Eick and Mawhinney [52] describe the same mechanism for a renormalization-group coarse lattice and need a Chebyshev filter of high degree; here $\tilde{A}_R$ is close to the identity on the high modes, and the smoother suffices.

$W(\tilde{A}_R \mathcal{M})$ crosses the origin on every configuration in both setups: with stout smearing at $M_5 = 1$ (setup S) and without it at $M_5 = 1.8$ (setup B). Of the three cases we ran, only at $M_5 = 1$ without smearing is the amplification far smaller and $W(\tilde{A}_R \mathcal{M})$ in the right half-plane, where truncated GCR(k) is guaranteed to converge for every k with a fixed cycle [44].

2. Spectrum

The spectrum, unlike the field of values, stays far from the origin (Table 17, Fig. 7). Every eigenvalue lies at $\mathrm{Re}\,\lambda \geq 0.70$, with moderate condition numbers. A few eigenvalues lie farther from 1 than the rest, isolated outliers (Table 17 gives those of configuration 1), and all other eigenvalues lie in a disc about 1 that excludes the origin. The left and right forms have nearly the same spectrum.

3. Pseudospectra and a bound for GMRES

On 12^4 (configuration 1) the pseudospectra of $\tilde{A}_R \mathcal{M}$ bound full GMRES with a fixed cycle: $\|r_k\|/\|r_0\| < 10^{-9}$ within 55 (S) and 57 (B) iterations (Table 18). For a non-normal operator the

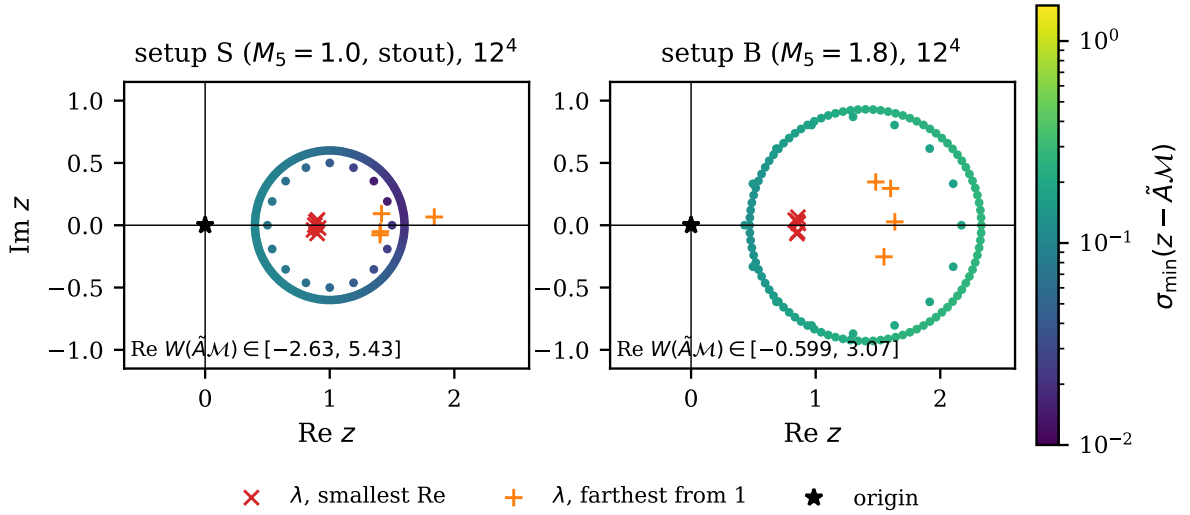


FIG. 7. 12^4, $m = 0.001$, exact coarse solves (the Pauli–Villars factor regularized): the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the contours of Table 18, among them the dense circles $|z - 1| = 0.6$ (S) and $|z - 1.4| = 0.93$ (B). The field of values reaches into the left half-plane (its real range in the corner), the spectrum does not.

convergence of GMRES is bounded by the ε-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$ [35, 50]. If a contour Γ of length L encloses the spectrum of X but not the origin and $\sigma_{\min}(z - X) \geq \varepsilon$ on it, then for any polynomial p_k with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \|p_k(X)\| \leq \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \tag{5.1}$$

An isolated outlier λ_o outside Γ is removed by the factor $(1 - z/\lambda_o)$ of p_k: then $p_k(X) = p_k(X)(I - P_o)$ with P_o the spectral projector, $\|(z - X)^{-1}(I - P_o)\| \leq \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and $p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \kappa_o \frac{r}{\varepsilon} \frac{|\lambda_o - c| + r}{|\lambda_o|} \left(\frac{r}{c}\right)^{k-1}. \tag{5.2}$$

On 16^4, configuration 1, setup S has two outliers; the factor κ_o is then replaced by $\|I - P_1 - P_2\| \leq 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the dense circles of 12^4 the spacing of the points is small enough that, $\sigma_{\min}(z - X)$ being 1-Lipschitz in z, the bound holds on the whole circle, to the accuracy of the Lanczos estimates.

On the dense circles the bound gives convergence at a fixed rate, r/c per iteration, with no stagnation; on the contours with 16 points it is an estimate. Its rate on the dense circle of setup S, $r/c = 0.6$, is close to the one we measure on $24^3 \times 48$, a larger lattice, in setup S at $m = 0.001$ and $\alpha = 1$ on configuration 1, where the median factor per GCR step of $|r|$ is 0.64 (Fig. 8). The Lanczos estimates approach $\sigma^2_{\min}$ from above, the disc rests on the eigenvalues farthest from c found by Arnoldi, and the bound is for full GMRES with a fixed $\mathcal{M}$.

4. Truncated GCR and inexact coarse solves

Truncated GCR does not need accurate coarse solves. With the usual coarse solves to 10^{-2}, GCR(2) needs in the median at most one iteration more than GCR(32) on 12^4 (Table 19). Coarse solves to 10^{-8}, where $\mathcal{M}$ is close to the fixed map, do not help: GCR(2) then needs 19 iterations and GCR(≥ 16) 15.

In the directions that E amplifies, the solver's cycle, with coarse solves to 10^{-2}, does better than the fixed map. At the vector x at the left end of $W(\tilde{A}_R\mathcal{M})$, coarse solves to 10^{-2} reduce $\|(I - \tilde{A}_R E)x\|$

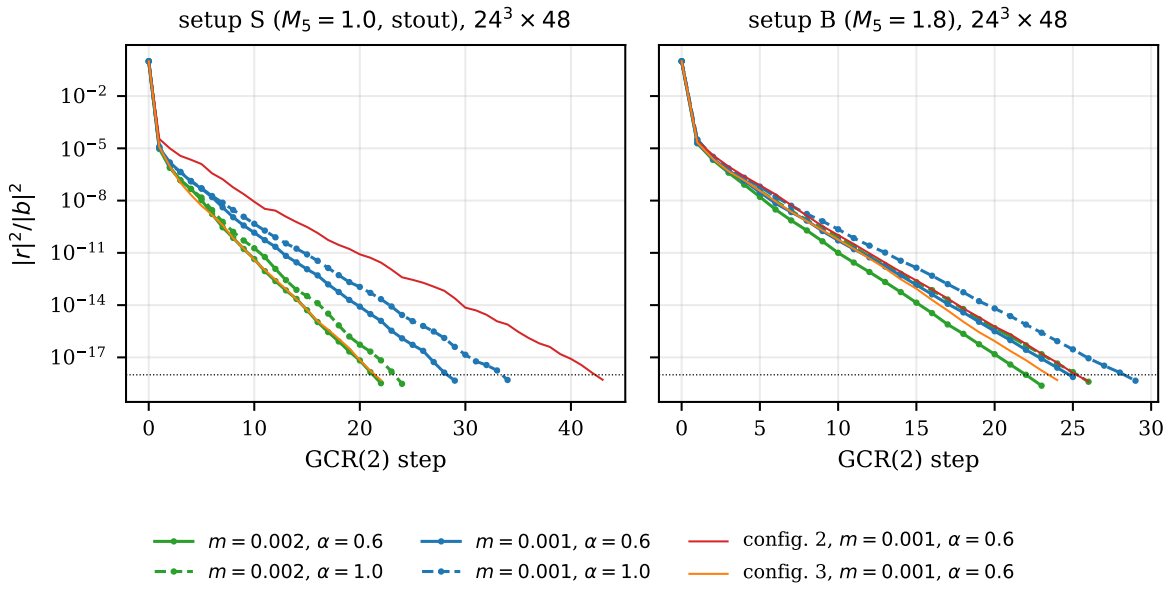


FIG. 8. GCR(2) residual histories on $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 4: configuration 1 at $m = 0.001$ (blue) and 0.002 (green), $\alpha = 0.6$ (solid) and 1 (dashed); configurations 2 (red) and 3 (orange) at $m = 0.001$, $\alpha = 0.6$.

by large factors against exact ones and in most cases make $\mathrm{Re}\, x^\dagger \tilde{A}_R \mathcal{M} x$ positive. With Krylov coarse solves stopped early the cycle is not a fixed linear map, so this holds for these vectors only.

A truncated method has no guarantee when the field of values contains the origin [44, 51]. With a fixed cycle GCR(2) can reach a fixed point, where the new direction is orthogonal to the residual and the next step repeats it. With the parameters of Table 4 no solve with D reached a plateau. The one plateau we checked, at $\kappa_s = 10$ in setup B (Sec. V.C.4), agrees with this picture, since the same cycle converges with GCR(16); we did not measure the angle between the new direction and the residual there.

Configuration 2 of $24^3 \times 48$ needs more iterations than the other three (Table 9); which property of that configuration causes this is open.

VI. Summary

At $m = 0.001$ on four $24^3 \times 48$ quenched configurations, one GPU, the multigrid solve is 4.1–7.3 times faster than mixed-precision even-odd CGNE, and a point-source propagator takes 0.40–0.56 of its wall time, setup included. Its iterations grow by a factor of 2.8–5.4 from $m = 0.1$ to $m = 0.001$, those of CGNE by 20–32. In solve time, setup excluded, the multigrid is faster from 16^4 on. BiCGStab and GCR(2) with the Chebyshev smoother on $\tilde{A}_L$ alone are slower than mixed-precision CGNE, so the gain comes from the coarse correction.

Each of the three reasons of Boyle and Yamaguchi is addressed by one choice: the even-odd system, a four-dimensional coarse lattice with the whole fifth dimension in each aggregate, and levels below the coarse lattice in place of deflation; the effect of the levels was not measured on its own. The lift $\hat{t}$ makes a given number of vectors y suffice: without it GCR(2) stagnates on some configurations of setup S or takes several times the iterations there, and it costs iterations in setup B, although the correction is exact only for the right form. Twice as many vectors y with s alone need fewer iterations on $24^3 \times 48$, at 1.6–1.7 times the setup time. One setup serves every value of Neff's α.

The field of values of the preconditioned operator reaches into the left half-plane, so the bound for GCR is lost in both setups. At $\alpha = 1$ its spectrum has $\mathrm{Re}\,\lambda \geq 0.70$, and on 12^4 (one configuration) its pseudospectra bound full GMRES with a fixed cycle. With the parameters of Table 4, truncated GCR(2) solved with D without a plateau on every volume, but it has no guarantee.

Solving with $D^\dagger$, as HMC needs, uses the cycle of D between two Γ_5, with no second setup. On 16^4 at $m = 0.01$ it takes one to four iterations more than solving with D; on 12^4 in setup S the residual right-hand side needs GCR(8) with two cycles per step, which converges on all four configurations. HMC itself is not implemented here.

The evidence is four quenched configurations per volume up to $24^3 \times 48$ on one GPU, compared with CGNE and with single-grid Krylov solvers only. A comparison with HDCG and with the coarsening of $D^\dagger_{\mathrm{PV}} D$ in one code, larger volumes and HMC are the next steps (Sec. VII). Applying the coarse operators to the 12 columns of a point source together should raise the arithmetic intensity from about one flop per byte to about ten, as estimated for Wilson-clover multigrid [42]; we have not measured this.

Code. All results were obtained with RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, a rewrite of Bridge++ [33, 34] in Rust whose GPU backend uses the CUDA kernels of the Wilson and Möbius operators of Bridge++. Where Bridge++ computes the same quantity, the results of RutticeQCD are compared with it number by number: the plaquette, the Wilson, clover and domain-wall operators applied to a vector, solver iteration counts, meson correlators, and single HMC trajectories. The domain-wall multigrid, which Bridge++ does not have, is checked against CGNE, whose propagators it reproduces (Sec. V.B.1). Its parameters in Table 4, k included, are inputs of the program.

VII. Open problems

The following questions remain open; the left form and the field of values are discussed in Secs. III.A.3, II.A and V.D.1–V.D.3.

1. **Parameters from the operator.** The smoother interval $[\lambda_{\mathrm{hi}}/\kappa_s, \lambda_{\mathrm{hi}}]$ uses a ratio chosen per setup, and a wrong one makes the solve stagnate (Sec. V.C.4); the setup works with singular values of $\tilde{A}_R$, which for a non-normal operator need not match its eigenvalues. How to set the interval from the operator, whether an ellipse with complex foci should replace the real interval, and why setup S needs more test vectors than setup B remain to be understood (Sec. V.C.4). The low modes may vary in the fifth dimension more than vectors not split in s can follow (Eick and Mawhinney [52] find their fifth-dimensional profiles to differ between lattices); a test vector split in s, as it is now split in chirality, would keep s and $\hat{t}$ in the range of $\mathcal{P}$, and Lüscher's coherence test would measure what the coarse space captures. Whyte, Stathopoulos and Romero [54] truncate the test vectors of Wilson fermions by a singular value decomposition on each aggregate; here the truncated basis would no longer contain s and $\hat{t}$ exactly, and whether the gain outweighs that loss is open.

2. **Convergence theory.** The pseudospectral bound rests on Lanczos estimates of $\sigma_{\min}$, which approach it from above, and on the eigenvalues farthest from the center found by Arnoldi (Sec. V.D.3). Obtaining a bound from an inclusion theorem, and tracking how the outliers of setup S and the pseudospectra develop beyond 16^4, remain to be done; on 24^4 the Arnoldi iteration does not fit in 24 GB of device memory. No theory covers truncated GCR with a preconditioner that varies from step to step. GCR(2) can reach a residual orthogonal to its new direction and stay there (Sec. V.D.4), and on the residual right-hand side of HMC the solve with $D^\dagger$ on 12^4 in setup S needs more kept directions (Sec. V.B.2); the solves with $D^\dagger$ were tested on 12^4 and 16^4 only. The cycle with D and D_{PV} exchanged and its Pauli–Villars factor regularized can also precondition solves with $D^\dagger$ from the same $\mathcal{P}$, with coarse operators of its own; it converges on 16^4 and stagnates with GCR(2) on 12^4 in setup S, and its cost in HMC remains to be measured. Why stout smearing gives the Galerkin Pauli–Villars factor singular values far below those of the fine operator (Sec. V.C.3) is not understood. We find that the field of values of the fine D_{PV} contains the origin with and without smearing and reaches further left with it, also on 16^4; the field of values alone does not explain the small singular values. A safeguard that changes the direction when $|\langle r|\tilde{A}\mathcal{M}r\rangle|/(\|r\|\,\|\tilde{A}\mathcal{M}r\|)$ falls below a threshold would remove such fixed points for one inner product per step; we have not

implemented it. A quantitative explanation for why $N = c[Y, W]D^{-1}$ has a small square, beyond the heuristic of Sec. IV, is also lacking.

3. **Formulations and coarse spaces.** The comparisons here are with CGNE and with Krylov solvers on $\tilde{A}_L$ alone. How the two nine-point factors compare with the 81-point HDCG operator and with the 33-point coarsening of $D^\dagger_{\mathrm{PV}} D$ [43], in cost per iteration and in iterations, needs all of them in one code on the same lattices.

Changes

2026-11-08. New Sec. IV.D: an exact solve with $D^\dagger$ through one solve with D, Eq. (4.8), and its comparison with $\Gamma_5 \mathcal{M} \Gamma_5$. Sec. V.A, footnote: Boriçi's operator, which needs a narrower smoother interval in setup S. Appendix D: the even-odd equation of CGNE with Neff's α is $\mathcal{K}^{(\alpha)} y = Q_e^{-1} C_e^{-1} \eta'_e$; the factor Q_e^{-1} was missing there and in b (the code solves the corrected equation). Sec. IV.C: a spectrum that surrounds the origin does not stop a Krylov method; it forces a degree of the order of the number of eigenvalues. Algorithms 7 and 5: the checks for an exact solution before ω is computed and before the Arnoldi vector is normalized.

Acknowledgments

I thank Hideo Matsufuru and Issaku Kanamori for many discussions on the multigrid solver. RuthiceQCD was written with reference to Bridge++ [33, 34]; its structure follows Bridge++ closely, and its correctness rests on a number-by-number comparison with it. I am also grateful to my colleagues at my current company for teaching me the practical details of algebraic multigrid in computational fluid dynamics, which inspired the two-lift prolongation $\{s, \hat{t}\}$. Although computational fluid dynamics and lattice QCD lie far apart in physics, their numerical methods have much to learn from one another. Finally, I thank my friend Sakura Itatani for helping me find a reference on local coherence. I used Claude (Anthropic) to assist with the code; I take full responsibility for the content.

TABLE 6. The notation of this work and that of other references. Equation numbers are those of the references; "the same" means the same operator.

	This work	Ref. [17]	Ref. [45]	Bridge++ code
Height	$M_5 > 0$	$-M_5$, $M_5 = O(-1)$ (2.13)	M_0	M_0, `domain_wall_height`
Fifth dim.	$s = 0, \ldots, L_s - 1$	$s = 1, \ldots, L_s$	$s = 1, \ldots, N_s$	$0, \ldots, N_s - 1$, `extent_of_5th_dimension`
Wilson part	$W = (4 - M_5) - \frac{1}{2}K$	$D^{\mathrm{Wilson}}(M_5)$ (2.13)	$D_{\mathrm{W}}(-M_0)$	$\frac{1}{2\kappa}D_{\mathrm{W}}$, $\kappa = 1/(8 - 2M_0)$
$D_\pm$	$D_+ = 1 + bW$, $D_- = cW - 1$ (Eq. (2.3))	$D_+ = bD^{\mathrm{W}} + 1$, $D_- = cD^{\mathrm{W}} - 1$ (2.18)	the same (2)	n/a
D	Eq. (2.8)	$D_{\mathrm{DW}}(m)$ (2.20)	$D(m)$ (1)	`Fopr_Domainwall`, mode `D`
C	$d_+ + d_- Y$, Eq. (2.15)	n/a	$\mathrm{diag}(D_{ee}, D_{oo})$ (4)	D_{ee}, D_{oo} of `_eo`; `m_dp` $= \alpha d_+$, `m_dm` $= \alpha d_-$
Neff's α	DQ, Eq. (2.22)	n/a (its $\alpha = \frac{b+c}{b-c}$)	n/a	`parameter_alpha`: DQ, site part CQ
Γ_5	$\gamma_5 R$	$\gamma_5 R D_-^{-1}$ (2.22)	n/a	n/a
4d quark	$P_-\psi_0 + P_+\psi_{L_s-1}$	$P_-\Psi_1$, $P_+\Psi_{L_s}$	n/a	the same

Appendices

A. The notation of other references

Table 6 relates Eqs. (2.6)–(2.22) to Brower, Neff and Orginos [17], to Kanamori, Chen and Matsufuru [45] and to the code of Bridge++ [34] (`Fopr_Domainwall`, `Fopr_Domainwall_eo`, with Neff's α as `parameter_alpha`). Eq. (2.4) is the tridiagonal matrix of Ref. [17], Eq. (2.20), and of Ref. [45], Eq. (1), with $D_+ = 1 + bW$ on the diagonal, $D_- P_+$ below and $D_- P_-$ above it, $-mD_- P_\pm$ in the corners, and s shifted by one. The operators D, C and $D^{(\alpha)}$ are thus the same in all four; the differences are names and signs:

- $D_\pm$ are those of Refs. [17, 45], $D_- = cW - 1$. The hop Y of Eq. (2.6) carries a minus sign, so $D = D_+ - D_- Y$ and $d_- = 1 - c(4 - M_5)$ is minus the site part of D_-.
- The height $M_5 > 0$ here is M_0 in Ref. [45] and in Bridge++ (`domain_wall_height`); Ref. [17] writes M_5 for the Wilson mass, $M_5^{\mathrm{BNO}} = -M_5$, Eq. (2.13). The caption of Fig. 1 of Ref. [45] gives $M_0 = -1$ for the height 1 of its text.
- The α of Ref. [17], $(b + c)/(b - c)$, is the scale factor of the Möbius kernel (2 at $b = 1.5$, $c = 0.5$), unrelated to Neff's α of Eq. (2.22).
- $\Gamma_5 = \gamma_5 R D_-^{-1}$ in Ref. [17], Eq. (2.22); here $\Gamma_5 = \gamma_5 R$.

$D^{(\alpha)}$ here and the operator of Bridge++ with `parameter_alpha` were compared on a 12^4 configuration with stout links, $L_s = 8$, $m = 0.01$: at $\alpha = 0.6$ and $\alpha = 1$ CG on the even-odd normal equation took the same number of iterations for every column of a point source, and the pseudoscalar correlators agree. Bridge++ returns $\psi^{(\alpha)}$; $\psi = Q\psi^{(\alpha)}$.

B. Spectra with other preconditioners

Figure 9 reproduces Fig. 3 of Ref. [45]. It shows the spectra that Sec. II.A compares with that of A': the unpreconditioned D, the site-diagonal DC^{-1}, the Pauli–Villars form $D_{\mathrm{PV}}^\dagger D$ of Ref. [41], and its block-diagonal approximations.

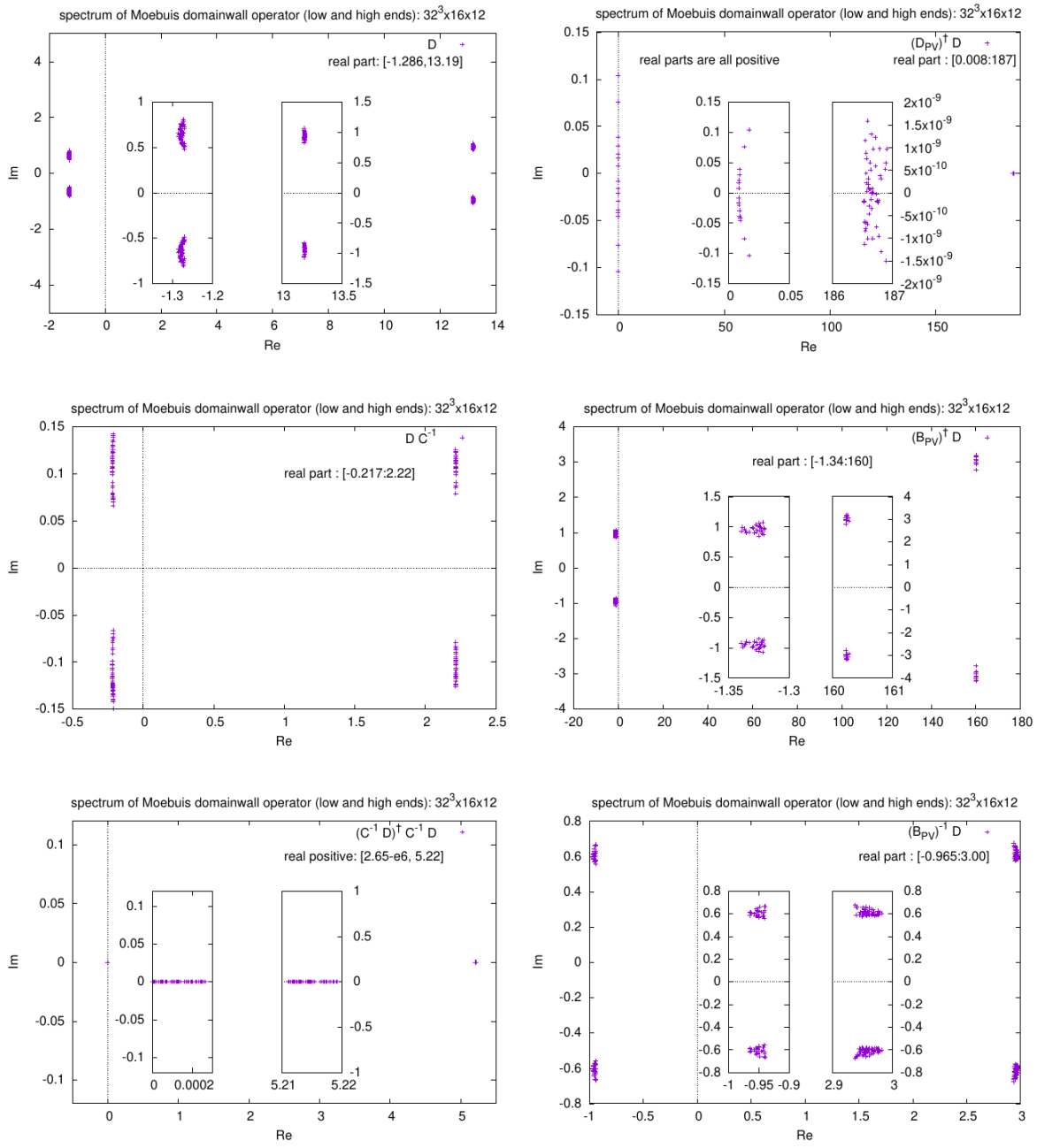


FIG. 9. Low and high ends of the spectrum of the Möbius operator on the configuration of Fig. 2, with preconditioners. Left column: D, DC^{-1}, $(C^{-1}D)^\dagger C^{-1}D$. Right column: $D^\dagger_{\rm PV}D$, $B^\dagger_{\rm PV}D$, $B^{-1}_{\rm PV}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). From Ref. [45], Fig. 3.

TABLE 7. The gauge configurations: plaquette, t_0/a^2, and the topological charge at t_0. The logs and the generation histories are listed with the data.

lattice		plaquette	t_0/a^2	Q_{clover}	Q_{imp}	Q
12^4	1	0.594250	3.19	+0.94	+1.00	+1
	2	0.593201	3.29	+0.95	+1.00	+1
	3	0.594204	3.78	+1.78	+1.96	+2
	4	0.594103	4.04	+1.88	+1.99	+2
16^4	1	0.593509	3.54	+0.05	+0.01	0
	2	0.593847	3.55	+0.95	+1.00	+1
	3	0.594395	3.18	+0.96	+1.00	+1
	4	0.593685	3.03	+2.53	+2.88	+3
24^4	1	0.593524	3.65	+3.63	+3.92	+4
	2	0.593789	3.41	+1.77	+1.94	+2
	3	0.593297	3.16	−3.55	−3.93	−4
	4	0.593786	3.24	−4.69	−5.05	−5
$24^3 \times 48$	1	0.593548	3.20	+4.47	+4.78	+5
	2	0.593630	3.18	+7.83	+8.42	+8
	3	0.593689	3.19	+3.46	+3.53	3 or 4
	4	0.593395	3.23	−1.57	−1.87	−2

C. Gauge configurations

All quenched configurations use the plaquette gauge action at $\beta = 6.0$ and were generated by HMC from a cold start, 400 trajectories of length 0.5 (on $24^3 \times 48$, 400 from a cold start and then 200); these are configurations 1. Configurations 2–4 continue the same chains with trajectories of length 0.5 and are taken 100 trajectories apart, after 502, 602 and 702 trajectories (702, 802 and 902 on $24^3 \times 48$). Table 7 gives the plaquette, the scale t_0 ($t^2E = 0.3$, E of the clover field strength) and the topological charge at t_0 after the Wilson flow (step 0.01), of the clover field strength and improved with rectangles ($c_{\text{plaq}} = 5/3$, $c_{\text{rect}} = -1/12$). On the larger lattices the two definitions differ by up to 1 at t_0.

D. Krylov solvers

The multigrid uses four Krylov solvers, flexible GCR for the outer system, BiCGStab and CG for the coarse factors and FGMRES in the bootstrap, and the comparison uses CG as well. In the algorithms below ϵ is the tolerance on $|r|^2/|b|^2$.

Flexible GCR(k). The outer solver of Sec. III.E is GCR [44] with a preconditioner $\mathcal{M}$ that may change from one step to the next, keeping the last k directions (Algorithm 4). Each step applies $\mathcal{M}$ once, makes the new $q = \tilde{A}z$ orthogonal to the q of the kept directions, and minimizes the residual along it; the step length $\langle q|r\rangle$ vanishes when q is orthogonal to r, which is the fixed point of Sec. V.D.4. After $n_{\max}$ steps the true residual is recomputed and the solve restarts.

Algorithm 4. Flexible GCR(k) for $\tilde{A}x = b$ with the preconditioner $\mathcal{M}$.

repeat
$r \leftarrow b - \tilde{A}x$
for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon|b|^2$ **do**
$z \leftarrow \mathcal{M}r$; $\quad q \leftarrow \tilde{A}z$
for each of the last k pairs (z_i, q_i) **do**
$\gamma \leftarrow \langle q_i|q\rangle$; $\quad q \leftarrow q - \gamma q_i$; $\quad z \leftarrow z - \gamma z_i$
end for
$z \leftarrow z/|q|$; $\quad q \leftarrow q/|q|$; $\quad$ keep (z, q)
$\beta \leftarrow \langle q|r\rangle$; $\quad x \leftarrow x + \beta z$; $\quad r \leftarrow r - \beta q$
end for
until $|b - \tilde{A}x|^2 < \epsilon|b|^2$

FGMRES. Each bootstrap round of Sec. III.C replaces a vector y_j by a few steps of flexible GMRES [23] on $\tilde{A}_R x = y_j$ from $x = 0$, with the cycle as a right preconditioner (Algorithm 5). The basis is orthogonalized by classical Gram–Schmidt applied twice, since one pass loses orthogonality on the strongly non-normal $\tilde{A}_R\mathcal{M}$, and the least-squares problem of the Hessenberg matrix is updated by Givens rotations at every step.

Algorithm 5. FGMRES(n) for $Ax = b$ with a right preconditioner $\mathcal{M}$.

repeat
$r \leftarrow b - Ax$; $\quad \beta \leftarrow |r|$; $\quad v_0 \leftarrow r/\beta$
for $k = 0, \ldots, n-1$, while the residual estimate is $\geq \epsilon^{1/2}|b|$ **do**
$z_k \leftarrow \mathcal{M}v_k$; $\quad w \leftarrow Az_k$
for two passes **do**
for $i = 0, \ldots, k$ **do** $h_{ik} \leftarrow h_{ik} + \langle v_i|w\rangle$; $\quad w \leftarrow w - \langle v_i|w\rangle v_i$
end for
end for
$h_{k+1,k} \leftarrow |w|$; **if** $h_{k+1,k} = 0$ **then** update the rotations and leave the loop (the solution lies in the space)
$v_{k+1} \leftarrow w/h_{k+1,k}$; $\quad$ update the Givens rotations and the residual estimate
end for
$c \leftarrow \arg\min_c |\beta e_0 - Hc|$; $\quad x \leftarrow x + \sum_i c_i z_i$
until $|b - Ax|^2 < \epsilon|b|^2$

CGNE. The reference solver of Sec. V.B.1 is CG on $\mathcal{K}^\dagger\mathcal{K}$ with $\mathcal{K} = C_e^{-1}S$, the even-odd operator of Bridge++, with Neff's α as for the multigrid, in double or in mixed precision.

For $D\psi = \eta$ the even-odd reduction of Sec. II.C gives $\mathcal{K}^{(\alpha)}y = Q_e^{-1}C_e^{-1}\eta'_e$ with $\mathcal{K}^{(\alpha)} = Q_e^{-1}\mathcal{K}Q_e$, Eq. (2.28), and CG solves its normal equation $Ay = b$, $A = \mathcal{K}^{(\alpha)\dagger}\mathcal{K}^{(\alpha)}$, $b = \mathcal{K}^{(\alpha)\dagger}Q_e^{-1}C_e^{-1}\eta'_e$; ψ_o and $\psi = Q\psi^{(\alpha)}$ follow as for the multigrid. The stopping criterion is $|b - Ay|^2/|b|^2 < 10^{-18}$. Since the

unknown is y itself and A has $\mathcal{K}^{(\alpha)}$ on the right, Bridge++ and Ref. [49] call this method CGNR; we keep the name CGNE that is common in lattice QCD.

Mixed precision. The mixed-precision solver runs the CG recurrence in single precision and corrects it by reliable updates [25, 26] in double precision (Algorithm 6): each time the single-precision $|\hat{r}|^2$ has fallen by a factor $\delta = 10^{-2}$ below its largest value since the last update, the single-precision solution is added to y and the true residual $b - Ay$ is recomputed in double precision; CG then goes on with the same search direction and stops only on the double-precision residual. With $\delta = 0$ the algorithm is plain CG in double precision; in single precision it is the solver of the Pauli–Villars factor, Eq. (3.9), started from $x = 0$. On $24^3 \times 48$ the mixed-precision solver needs the iterations of double precision within 1% except at $m \leq 0.002$ in setup S, where it needs up to 67% more (26% on configuration 1), and takes 0.53–0.88 times the time (over the four configurations), with the same final residuals and 7.7 GB of device memory against 7.4 GB in double precision.

Algorithm 6. CGNE with reliable updates. Hats: single precision; $\hat{A}$ is A on the single-precision links.

$r \leftarrow b - Ay$; **if** $|r|^2 < \epsilon|b|^2$ **return** y

$\hat{r} \leftarrow r$; $\quad \hat{p} \leftarrow \hat{r}$; $\quad \hat{x} \leftarrow 0$; $\quad \rho \leftarrow |\hat{r}|^2$; $\quad \rho_{\max} \leftarrow \rho$

loop

$\alpha \leftarrow \rho / \langle \hat{p} | \hat{A}\hat{p} \rangle$; $\quad \hat{x} \leftarrow \hat{x} + \alpha\hat{p}$; $\quad \hat{r} \leftarrow \hat{r} - \alpha\hat{A}\hat{p}$; $\quad \rho' \leftarrow |\hat{r}|^2$

if $\rho' < \delta\rho_{\max}$ **or** $\rho' < \epsilon|b|^2$ **then** ▷ reliable update

$y \leftarrow y + \hat{x}$; $\quad \hat{x} \leftarrow 0$; $\quad r \leftarrow b - Ay$

if $|r|^2 < \epsilon|b|^2$ **return** y

$\hat{r} \leftarrow r$; $\quad \rho' \leftarrow |\hat{r}|^2$; $\quad \rho_{\max} \leftarrow \rho'$

else

$\rho_{\max} \leftarrow \max(\rho_{\max}, \rho')$

end if

$\hat{p} \leftarrow \hat{r} + (\rho'/\rho)\,\hat{p}$; $\quad \rho \leftarrow \rho'$

end loop

BiCGStab. The coarse solves of the quark factor (Sec. III.B) use BiCGStab [22], Algorithm 7, with a right preconditioner $\mathcal{M}_A$. It bounds the angle between t and s from below as in Ref. [24], and after a breakdown or $n_{\max}$ steps it restarts from the true residual. The same algorithm, with $\mathcal{M}_A = 1$ and in double precision, is the Krylov solver on $\tilde{A}_L$ alone of Sec. V.C.1.

Algorithm 7. BiCGStab for $Ax = g$ with a right preconditioner $\mathcal{M}_A$; $\omega_c = 0.7$.

repeat
$r \leftarrow g - Ax$; $\quad r_0 \leftarrow r$; $\quad p \leftarrow r$; $\quad \rho \leftarrow \langle r_0 | r \rangle$
for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon |g|^2$ **do**
$\hat{p} \leftarrow \mathcal{M}_A p$; $\quad v \leftarrow A\hat{p}$; $\quad \alpha \leftarrow \rho / \langle r_0 | v \rangle$; $\quad s \leftarrow r - \alpha v$
if $|s|^2 < \epsilon |g|^2$ **then** $x \leftarrow x + \alpha \hat{p}$ and **return** x
$\hat{s} \leftarrow \mathcal{M}_A s$; $\quad t \leftarrow A\hat{s}$; $\quad \omega \leftarrow \langle t | s \rangle / |t|^2$
$\gamma \leftarrow |\langle t | s \rangle| / (|t|\,|s|)$; **if** $\gamma < \omega_c$ **then** $\omega \leftarrow \omega\, \omega_c / \gamma$
$x \leftarrow x + \alpha \hat{p} + \omega \hat{s}$; $\quad r \leftarrow s - \omega t$; $\quad \rho' \leftarrow \langle r_0 | r \rangle$
$p \leftarrow r + (\rho'/\rho)(\alpha/\omega)\,(p - \omega v)$; $\quad \rho \leftarrow \rho'$
end for
until $|g - Ax|^2 < \epsilon |g|^2$

Two even-odd forms. Boyle and Yamaguchi [40] used CG on $S^\dagger S$, which for Shamir fermions needed 3224 iterations against 3880 for the left-preconditioned form. Fig. 10 compares the two on the three quenched lattices at $m = 0.001$, in double precision with one random right-hand side, for $b - c = 1$ and five values of c; all other parameters are those of setups S and B.

The two operators differ by a site-local factor on the left, $S = C_e \mathcal{K}$, and with Neff's α, $SQ_e = C_e Q_e \mathcal{K}^{(\alpha)}$. Their condition numbers therefore differ at most by the condition number of that factor. Since $C = d_+ + d_- Y$ with $\|Y\| = 1$ (Eq. (2.8)), the singular values of C lie in $[d_+ - |d_-|,\, d_+ + |d_-|]$ and

$$\kappa(C) \leq \frac{d_+ + |d_-|}{d_+ - |d_-|}, \qquad d_+ = 1 + b(4 - M_5), \quad d_- = 1 - c(4 - M_5); \tag{D.1}$$

for $\alpha < 1$ the bound is $1/\alpha$ times larger, since $\kappa(Q) = 1/\alpha$. The measured ratios of the iterations stay within this range too, although the bound is on the condition numbers.

Within the bound the order depends on c. In setup B the ratio of the iterations grows smoothly with c, from 0.81–0.83 for Shamir fermions to 1.27–1.31 at $(b, c) = (2, 1)$, and crosses 1 near the $(b, c) = (1.5, 0.5)$ used in the main text. For Shamir fermions $S^\dagger S$ is the faster form by 1.20–1.23, close to the $3880/3224 = 1.20$ of Ref. [40]. In setup S the ratio is above 1 except at $c = 0.25$, where $S^\dagger S$ needs 3–5% fewer iterations; we have no explanation for this dip. The three lattices agree to within 0.05 in every case, so the order is a property of the operator and its parameters and does not depend on the configuration.

Neff's α favors $\mathcal{K}^\dagger \mathcal{K}$. With $\alpha = 0.6$, $S^\dagger S$ needs 46–58% more iterations in setup S and 14–32% more in setup B, for every c and on all three lattices, Shamir fermions included. In $\mathcal{K}^{(\alpha)}$ the factor $C_e Q_e$ is divided out on the left, whereas in SQ_e the rescaling acts on the right only. The singular values of the two operators would show whether this is the cause. For the setups of this work $\mathcal{K}^\dagger \mathcal{K}$ is the faster form, and the multigrid is compared with it.

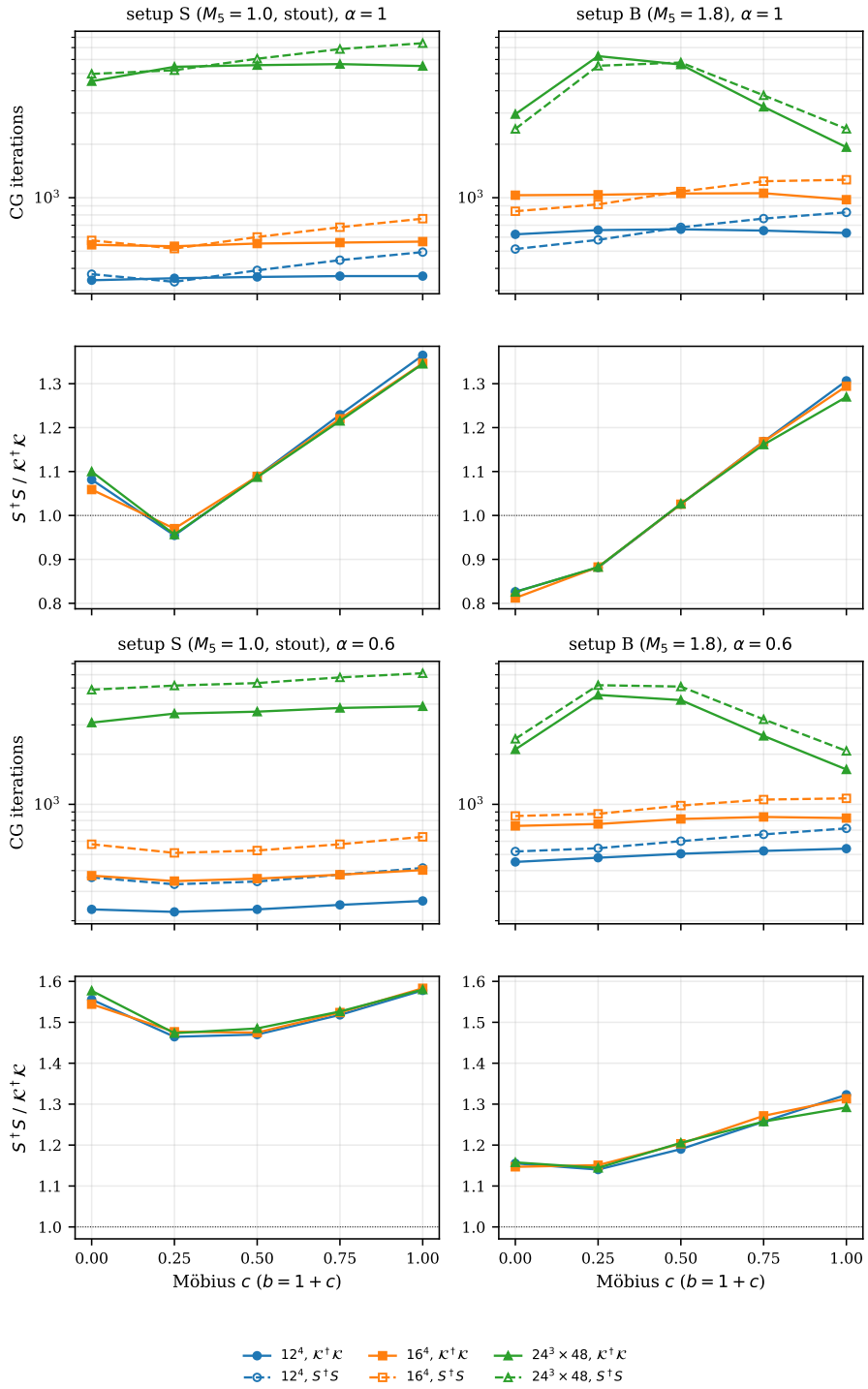


FIG. 10. CG iterations on $\mathcal{K}^\dagger\mathcal{K}$ (filled, solid) and on $S^\dagger S$ (open, dashed), with $\mathcal{K} = 1 - C_e^{-1} D_{eo} C_o^{-1} D_{oe}$ (the even-odd operator of Bridge++) and $S = C_e - D_{eo} C_o^{-1} D_{oe}$ (the Schur complement of Grid's default `SchurDiagMooeeOperator`, there on the odd sites), against the Möbius c, $b = 1 + c$, and below each their ratio; above 1 the left-preconditioned form is faster. Upper four panels $\alpha = 1$, lower four $\alpha = 0.6$. CG stops at $|r|^2/|b|^2 < 10^{-18}$ of its normal equations.

TABLE 8. Mass scan on $24^3 \times 48$, means over the four configurations, one random right-hand side to 10^{-18}, for CGNE on $|D\psi - \eta|^2/|\eta|^2$, for the multigrid on $|r|^2/|b|^2$ of its system: even-odd CGNE in double and in mixed precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations (for CGNE in double precision), solve time in seconds, coarse-operator applications, and N_{rhs}, the number of right-hand sides at which the multigrid setup (192–220 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$, against the double- and the mixed-precision CGNE (n/a: CGNE is faster); speed-up: the solve time of the mixed-precision CGNE over that of the multigrid, smallest and largest over the four configurations. Over the four configurations at $m = 0.001$, N_{rhs} against mixed CGNE is 2.6–4.8, and from $m = 0.1$ to 0.001 the iterations grow by 2.8–5.4 (MG) and 20–32 (mixed CGNE). Appendix D compares the two precisions of CGNE. Final $|D\psi - \eta|^2/|\eta|^2$: 1×10^{-21}–1×10^{-18} (CGNE, both), 2×10^{-19}–4×10^{-18} (MG). Device memory peak: CGNE 7.4 GB (double) and 7.7 GB (mixed), MG 21.1 GB ($\alpha = 0.6$).

	CGNE			MG, $\alpha = 0.6$			MG, $\alpha = 1$		N_{rhs}		speed-up
m	it	s	s, mixed	it	s	coarse	it	s	double	mixed	mixed
setup S ($M_5 = 1.0$, stout)											
0.1	199	6.9	4.5	8	3.3	713	11	4.5	54	161	1.3–1.5
0.05	342	10.9	6.6	11	4.3	851	14	5.4	29	83	1.5–1.6
0.02	761	22.2	12.7	12	4.6	879	14	5.5	11	24	2.7–3.0
0.01	1399	39.4	21.8	14	5.7	1248	15	6.4	5.9	12	3.6–4.0
0.005	2550	70.5	38.2	17	7.4	1825	19	8.5	3.2	6.5	4.4–6.0
0.002	3646	100.2	58.5	23	11.3	3213	26	12.3	2.3	4.3	4.6–5.9
0.001	3749	103.0	70.0	29	14.9	4585	31	16.0	2.4	3.8	4.1–5.4
setup B ($M_5 = 1.8$)											
0.1	251	8.2	5.3	7	3.2	303	10	4.5	41	101	1.6–1.7
0.05	455	13.9	8.3	10	4.6	437	13	5.7	22	55	1.8
0.02	1009	28.9	16.3	13	6.0	593	15	6.9	8.9	20	2.7–2.8
0.01	1739	48.6	26.7	15	7.0	730	18	8.3	4.9	10	3.7–4.0
0.005	2827	78.1	42.2	18	8.5	980	21	9.9	2.9	6.1	4.7–5.2
0.002	4446	121.8	65.3	22	10.5	1304	26	12.1	1.9	3.8	5.8–6.6
0.001	5490	150.1	80.3	25	11.8	1479	28	13.1	1.5	3.0	6.5–7.3

E. Tables of the numerical results

This appendix holds the tables that support Sec. V. Medians over an even number of configurations are the mean of the two middle values; halves are rounded up.

TABLE 9. Iterations and solve seconds at $m = 0.001$ against the volume, one random right-hand side: ranges over the four configurations; multigrid: plus its mean setup time. BiCGStab: configuration 1.

			12^4	16^4	24^4	$24^3 \times 48$
S	CGNE, mixed	it	240–326	369–1089	2124–3253	3829–6652
		s	0.2	0.7–1.9	16.3–24.5	56.4–96.7
	BiCGStab on $\tilde{A}_L$	it	235	398	1905	3247
		s	0.4	2.0	49.9	169.5
	multigrid	it	11–18	10–11	14–24	22–43
		s	0.4–0.7 (+7)	0.5–0.8 (+17)	2.9–5.3 (+96)	10.5–23.6 (+209)
B	CGNE, mixed	it	556–729	862–1630	3518–4543	5104–6281
		s	0.4–0.5	1.5–2.6	26.3–33.7	75.1–91.3
	BiCGStab on $\tilde{A}_L$	it	368	663	1909	2085
		s	0.7	3.4	50.0	108.8
	multigrid	it	12–13	13–14	23–24	23–26
		s	0.4 (+6)	0.8 (+18)	5.5–5.8 (+99)	11.1–12.5 (+205)

TABLE 10. GCR iterations of a solve with $D^\dagger$ to $|r|^2/|b|^2 < 10^{-10}$ from one setup of D (32 test vectors, blocks 4^4). "rand": random right-hand side; "res": the residual after one sweep of $D^\dagger D$ (Sec. V.B.2). Medians over the four configurations, in brackets the number that do not converge in 300 iterations ("stall": three or four; with two, the range of the other two). One inner cycle per step unless stated.

	16^4, $m = 0.01$				12^4, $m = 0.001$			
	S		B		S		B	
method	rand	res	rand	res	rand	res	rand	res
D, GCR(2)	5	7	7	9	5	7	7	9
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(2)	9	10	10	12	13	stall	12	14
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(8)	9	10	10	11	11	17 [1]	11	13
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(2), two cycles	9	9	10	9	11	13–15 [2]	10	13
$\Gamma_5\mathcal{M}\Gamma_5$, GCR(8), two cycles	9	9	9	9	10	10	10	11

TABLE 11. GCR(2) iterations on $\tilde{A}_L$ with $\mathcal{P}$ made of different lifts of the vectors y, $m = 0.001$, one random right-hand side; the other parameters as in Table 4; setup seconds on $24^3 \times 48$. 12^4 and 16^4: medians over the four configurations, in brackets the number that do not converge in 2000 iterations (with two, the range of the other two); $24^3 \times 48$: configuration 1.

			12^4		16^4		$24^3 \times 48$		setup (s)
lifts	y	columns	S	B	S	B	S	B	S, B
$\{s, \hat{t}\}$	16	32	11	13	11	14	29	25	204, 205
$\{s, Ds\}$	16	32	21 [1]	15	22	18	73	32	209, 205
$\{s\}$	16	16	23 [1]	15	12–44 [2]	16	92	32	138, 179
$\{s\}$	32	32	12	12	12	11	13	11	330, 340

TABLE 12. GCR(2) iterations on $\tilde{A}_L$ for seven setups of setup S on 12^4 (configuration; a–c: setups with different random seeds), $m = 0.001$: without the shift (BiCGStab on the exact adjoint), with $\tau = 10^{-3}$, and the range over $\tau = 10^{-4}, 3\times10^{-4}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}$; otherwise Table 4. n.c.: not converged in 2000 iterations. *One run, $\tau = 10^{-2}$: 63.

	no shift	$\tau = 10^{-3}$	other τ
1a	n.c.	15	11–13
1b	70	14	13–19
1c	270	14	13–14
2a	106	12	12–14
3a	n.c.	13	12–15
3b	204	13	11–14*
4, default seed	n.c.	18	14–16

TABLE 13. GCR(2) iterations on $\tilde{A}_L$ without and with the Tikhonov shift of the Pauli–Villars coarse solve, setup S, $m = 0.001$, default seed, configurations 1–4; without the shift the factor is solved by BiCGStab on the exact adjoint. n.c.: not converged in 2000 iterations.

		configuration			
		1	2	3	4
12^4, stout	no shift	610	84	883	n.c.
	$\tau = 10^{-3}$	11	11	11	18
16^4, stout	no shift	44	50	33	29
	$\tau = 10^{-3}$	10	11	11	11
12^4, no smearing	no shift	14	15	15	15
	$\tau = 10^{-3}$	13	13	14	13

TABLE 14. Smallest singular value of the coarse Pauli–Villars factor $\mathcal{D}_c^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ / that of the fine D_{PV} (Lanczos on $X^\dagger X$), setup S, $m = 0.001$, default seed. Stout: the smearing of setup S; none: the same setup without smearing. $\mathcal{P}$ comes from a setup whose cycle solves the Pauli–Villars factor without the shift, by BiCGStab on the exact adjoint as in Table 13.

	smearing	configuration 1	2	3	4
12^4	stout	0.0129 / 0.887	0.0120 / 0.497	0.0131 / 0.668	0.0036 / 0.435
12^4	none	0.259 / 0.436	0.237 / 0.393	0.236 / 0.406	0.232 / 0.402
16^4	stout	0.0151 / 0.692	0.0165 / 0.388	0.0258 / 0.656	0.0425 / 0.393

TABLE 15. The changes of Fig. 6 that move the cost, with timings: iterations / setup seconds / solve seconds, medians over the four configurations as in Fig. 6. 12^4, $m = 0.001$, $\tilde{A}_L$, GCR(2), one random right-hand side. n_v test vectors are the lifts $\{s_j, \hat{t}_j\}$ of $n_v/2$ vectors y_j (Sec. III.A.3).

change	setup S	setup B
base	11 / 7 / 0.4	13 / 6 / 0.4
aggregates 2^4	9 / 20 / 1.8	9 / 13 / 1.4
aggregates 3^4	10 / 8 / 0.5	11 / 7 / 0.4
aggregates 6^4	15 / 7 / 0.4	15 / 7 / 0.4
test vectors 16	15 / 4 / 0.3	18 / 4 / 0.3
test vectors 24	13 / 5 / 0.4	14 / 5 / 0.3
test vectors 48	9 / 15 / 0.7	11 / 11 / 0.6
smoother $\nu = 4$	15 / 6 / 0.5	stag.
smoother $\nu = 6$	12 / 6 / 0.4	45 [1] / 6 / 1.1
smoother $\nu = 12$	11 / 7 / 0.5	(base)
smoother $\nu = 16$	10 / 8 / 0.4	12 / 7 / 0.4
bootstrap rounds 1	31 [1] / 4 / 1.1	49 / 4 / 0.9
bootstrap rounds 3	14 / 10 / 0.7	12 / 9 / 0.4

TABLE 16. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R\mathcal{S}$, $\tilde{A}_R E$, $\tilde{A}_R\mathcal{M}$ (exact coarse solves, the Pauli–Villars factor regularized), $m = 0.001$, with the multigrid of the mass scan (16 low modes, 32 test vectors, 4^4, Chebyshev of Table 4) but no levels below the coarse lattice. Rows "4 configs." and "no smearing": ranges over the four 12^4 configurations ($M_5 = 1$ without smearing); other rows configuration 1. On 24^4 only $W(\tilde{A}_R)$ is given (Krylov dimensions 10 and 12 agree to six digits); there the field of values of $\tilde{A}_R\mathcal{M}$ did not settle (the two dimensions differ by 2%) and the eigensolvers did not fit in 24 GB of device memory.

	$\mathrm{Re}\, W(\tilde{A}_R)$	$\|I - \tilde{A}_R\mathcal{S}\|$	$\|I - \tilde{A}_R E\|$	$\mathrm{Re}\, W(\tilde{A}_R\mathcal{M})$	$\|I - \tilde{A}_R\mathcal{M}\|$
12^4 S	$[-4.59 \times 10^{-3}, 1.98]$	1.018	2.74×10^3	$[-2.63, 5.43]$	8.06
12^4 S, 4 configs.	≥ -0.013	1.02–1.13	2.1×10^3–3.2×10^3	≥ -3.11	6.9–9.3
12^4 S, no smearing	≥ -0.011	1.07–1.08	63–82	$[0.23, 2.58]$	1.5–2.0
12^4 B	$[-5.02 \times 10^{-2}, 4.46]$	1.112	1.84×10^3	$[-0.60, 3.07]$	3.65
12^4 B, 4 configs.	≥ -0.051	1.11–1.22	1.7×10^3–1.9×10^3	≥ -0.93	3.1–4.1
16^4 S	$[-5.07 \times 10^{-3}, 1.98]$	1.027	2.37×10^3	$[-1.53, 5.31]$	6.79
16^4 B	$[-5.06 \times 10^{-2}, 4.46]$	1.145	1.58×10^3	$[-1.04, 3.54]$	4.51
24^4 S	$[-1.31 \times 10^{-2}, 1.98]$	–	–	–	–
24^4 B	$[-5.08 \times 10^{-2}, 4.48]$	–	–	–	–

TABLE 17. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves, the Pauli–Villars factor regularized, $\alpha = 1$) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger \tilde{A}^\dagger$), the eigenvalues farthest from 1 with their κ, and the radius about 1 of the disc that holds all the others. $m = 0.001$. On 16^4 the smallest real part and κ are ranges over the four configurations, the outliers and the radius are those of configuration 1.

	$\min \operatorname{Re} \lambda$	κ (6 lowest)	outliers (κ)	others: $\lvert\lambda - 1\rvert \leq$
12^4 S, R	0.870	5.7–11.3	$1.84 + 0.07i$ (9.2)	0.426
12^4 S, L	0.862	6.4–14.2	$1.78 + 0.05i$ (10.1)	0.447
12^4 B, R	0.836	10.1–18.2	none	0.670
12^4 B, L	0.834	10.4–18.7	none	0.674
16^4 S, R	0.725–0.804	2.9–19.4	$2.92 + 0.06i$ (3.4)	0.453
			$1.93 + 0.00i$ (5.4)	
16^4 B, R	0.704–0.784	9.2–33.8	none	0.745

TABLE 18. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (exact coarse solves, the Pauli–Villars factor regularized), configuration 1, on circles $|z - c| = r$ (16 points each, or the number given), and the bound (5.1) or (5.2): the factor per iteration r/c, the constant, and the iterations it gives (a bound on the dense circles only) for $\|r_k\|/\|r_0\| < 10^{-9}$. For the dense circles ε is the lower bound $\min_i \sigma_{\min}(z_i) - h/2$ on the whole circle, h the spacing of the points ($\sigma_{\min}$ is 1-Lipschitz in z); otherwise the smallest sampled value. Lanczos to 10^{-3} relative in σ^2.

	c	r	$\min \sigma_{\min}$	r/c	constant	k for 10^{-9}
12^4 S, 160 points	1	0.6	0.0179, ≥ 0.0061	0.6	704	55
12^4 S	1	0.5	0.0146	0.5	229	39
12^4 B, 96 points	1.4	0.93	0.1222, ≥ 0.0918	0.66	10.1	57
12^4 B	1.3	0.87	0.1382	0.67	6.3	57
16^4 S	1	0.6	0.0326	0.6	123	52
16^4 S	1	0.55	0.0282	0.55	123	45
16^4 B	1.4	1.05	0.1189	0.75	8.8	80
16^4 B	1.45	1.1	0.1189	0.76	9.3	84

TABLE 19. GCR(k) iterations to $|r|^2/|b|^2 < 10^{-18}$, 12^4, $m = 0.001$, the multigrid of Table 4 (no levels below the coarse lattice), one random right-hand side: medians over the four configurations, in brackets the number that do not converge in 1000 iterations; the point source: range over its 12 columns and the configurations. Empty cells: not run.

	coarse solves	$k = 2$	8	16	32	64
S, $\tilde{A}_L$	10^{-2}	11			11	
S, $\tilde{A}_R$	10^{-2}	12			11	
B, $\tilde{A}_L$	10^{-2}	13			12	
B, $\tilde{A}_R$	10^{-2}	15			15	
S, $\tilde{A}_L$, point source	10^{-2}	11–19			11–13	
S, $\tilde{A}_L$	10^{-8}	19	16	15	15	15
S, $\tilde{A}_L$	10^{-4}	13			12	

TABLE 20. GCR(2) iterations on $\tilde{A}_L$ against the number of test vectors, $m = 0.001$, one random right-hand side, otherwise Table 4: ranges over the four configurations; n.c. (k): k configurations not converged in 2000 iterations. n_v test vectors are the lifts $\{s_j, \hat{t}_j\}$ of $n_v/2$ vectors y_j (Sec. III.A.3). 32 is the multigrid of the other tables (Tables 9 and 8).

setup	test vectors	24^4	$24^3 \times 48$
S	16	n.c. (4)	274, n.c. (3)
S	24	68–101, n.c. (1)	90–179
S	32	14–24	22–43
S	48	9–10	not run
B	16	72–79	65–83
B	24	40–42	38–45
B	32	23–24	23–26
B	48	11	not run

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